Exercise C - II

1. A variable plane is at a constant distance p from the origin and meets the coordinate axes in points A, B and C respectively. Through these points, planes are drawn parallel to the coordinates planes. Find the locus of their point of intersection.

एक चर तल जो कि मूल बिन्दु से नियत दूरी p पर स्थित है तथा निर्देशी अक्षों को बिन्दुओं A, B तथा C पर मिलती है। इन बिन्दुओं से निर्देश तलों के समानान्तर खीचे गये तलों के प्रतिच्छेद बिन्दु का बिन्दुपथ ज्ञात कीजिए।

Ans. $\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}=\frac{1}{p^2}$

Sol. Let plane is $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$ where a, b, c are intercepts on x–axis, y–axis and z–axis.

distance from origin

Given $d_{(0,0,0)} = P$

$$\frac{1}{\sqrt{\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}}}=P$$

$$\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}=\frac{1}{P^2}$$

Planes parallel to co-ordinate plane

$x = a, y = b, c = z$

POI of these planes (a, b, c)

Locus of POI

$$\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}=\frac{1}{P^2}$$

$$\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}=\frac{1}{P^2}$$

2. Let $\Pi : x + y - z - 4 = 0$ be the equation of a plane and A be the point with position vector $\hat{i}+2\hat{j}-3\hat{k}$. L is a line which passes through the point (1, 2, 3) with direction ratios 3, – 1 and 4. If the distance of the point A from the line L measured parallel to the plane Π is d_1 and the distance of the point A from the plane Π measured parallel to the line L is d_2, then find the value of $\sqrt{d_1^2 - d_2^2}$.

माना $\Pi : x + y - z - 4 = 0$ एक तल का समीकरण है तथा A एक बिन्दु है जिसका स्थिति सदिश $\hat{i}+2\hat{j}-3\hat{k}$ है। L एक रेखा है जो बिन्दु (1, 2, 3) से गुजरती है तथा जिसके दिक् अनुपात 3, – 1 तथा 4 है। यदि बिन्दु A की रेखा L से तल Π के समान्तर नापी गई दूरी d_1 है तथा बिन्दु A की तल Π से रेखा L से समान्तर नापी गई दूरी d_2 है तो $\sqrt{d_1^2 - d_2^2}$ का मान ज्ञात कीजिये।

Ans. 10

Sol. $L: \frac{x-1}{3}=\frac{y-2}{-1}=\frac{z-3}{4}=r$

$A: (1, 2, -3)$; $x + y - z = 6$

Any point say B on line is

$3r + 1, 2 - r, 4r + 3$

$\overrightarrow{AB} = (3r)\hat{i} + (-r)\hat{j} + (4r+6)\hat{j}$

$\vec{n} = \hat{i} + \hat{j} - \hat{k}$

$\overrightarrow{AB} \cdot \vec{n} = 0 \Rightarrow 3r - r - 4r - 6 = 0$

$2r = -6 \Rightarrow r = -3.$

Hence B is $(-8, 5, -9)$

$AP = \sqrt{81+9+36} = \sqrt{126} \Rightarrow d_1^2 = 126.$

Again line through $(1, 2, -3)$ with dr's $3, -1, 4$

$\frac{x-1}{3}=\frac{y-2}{-1}=\frac{2+3}{4}=\lambda$

$\left.\begin{array}{l} x = 3\lambda + 1 \\ y = 2 - \lambda \\ z = 4\lambda - 3 \end{array}\right]$ must satisfy $x + y - z = 4.$

$3\lambda + 1 + 2 - \lambda - 4\lambda + 3 = 4$

$-2\lambda + 6 = 4 \Rightarrow \lambda = 1$

Hence the point on the plane is $(4, 1, 1)$

$\therefore$ Distance of A from the plane measured parallel to the line $= \sqrt{9+1+16} = \sqrt{26}$

$\Rightarrow$ $d_2^2 = 26$

Hence $\sqrt{d_1^2 - d_2^2} = \sqrt{126-26} = \sqrt{100} = 10.$ Ans.

3. The position vectors of the four angular points of a tetrahedron OABC are (0, 0, 0); (0, 0, 2); (0, 4, 0) and (6, 0, 0) respectively. A point P inside the tetrahedron is at the same distance 'r' from the four plane faces of the tetrahedron. Find the value of 'r'.

चतुष्फलक OABC के चारो कोणीय बिन्दुओं के निर्देशांक क्रमशः (0, 0, 0); (0, 0, 2); (0, 4, 0) तथा (6, 0, 0) चतुष्फलक के अन्दर चारों तलीय फलकों से समदूरस्थ (दूरी 'r') बिन्दु P है। r का मान बताइये।

Ans. $\frac{2}{3}$

Sol. Angular point OABC are (0, 0, 0), (0, 0, 2), (0, 4, 0) & (6, 0, 0)

Let centre of sphere be (r, r, r)

Equation of plane passing ABC is

$$\frac{x}{6}+\frac{y}{4}+\frac{z}{2}=1$$

$$r=\left|\frac{\frac{r}{6}+\frac{r}{4}+\frac{r}{2}-1}{\sqrt{\frac{1}{6^2}+\frac{1}{4^2}+\frac{1}{2^2}}}\right|$$

$7r = \pm (11r - 12)$

$r=\frac{2}{3}$, $r = 3$ (not satisfied)

4. The line $\frac{x+6}{5}=\frac{y+10}{3}=\frac{z+14}{8}$ is the hypotenuse of an isosceles right angled triangle whose opposite vertex is (7, 2, 4). Find the equation of the remaining sides.

रेखा $\frac{x+6}{5}=\frac{y+10}{3}=\frac{z+14}{8}$ एक समद्विबाहु समकोण त्रिभुज का विकर्ण है। जिसके समाने का शीर्ष (7, 2, 4) है। बाकी भुजाओं का समीकरण ज्ञात कीजिए।

Ans. $\frac{x-7}{3}=\frac{y-2}{6}=\frac{z-4}{2}$; $\frac{x-7}{2}=\frac{y-2}{-3}=\frac{z-4}{6}$

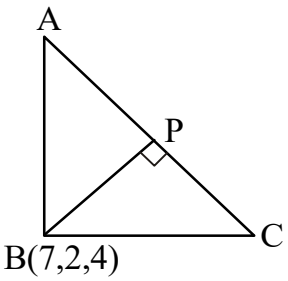
Sol. $P = (5k - 6, 3k - 10, 8k - 14)$

Now $\overrightarrow{BP}\cdot\overrightarrow{AC}=0$

$5(5k - 13) + 3(3k - 12) + 8(8k - 18) = 0$

$k=\frac{245}{98}=\frac{5}{2}$

$P=\left(\frac{13}{2},-\frac{5}{2},6\right)$, Now length $BP=\frac{\sqrt{98}}{2}$



Now $A,C=\left(\frac{13}{2}\pm\frac{5}{\sqrt{98}}.\frac{\sqrt{98}}{2},-\frac{5}{2}\pm\frac{3}{\sqrt{98}.}\frac{\sqrt{98}}{2},6\pm\frac{8}{\sqrt{98}},\frac{\sqrt{98}}{2}\right)$

$A = (9, -1, 10)$

$C = (4, -4, 2)$

Equation of BC $\frac{x-7}{3}=\frac{y-2}{6}=\frac{z-4}{2}$

Equation of AC $\frac{x-7}{2}=\frac{y-2}{-3}=\frac{z-4}{6}$

5. Consider a plane passing through three points A(a, 0, 0), B(0, b, 0), C(0, 0, c) with a > 0, b > 0, c > 0. Let d be the distance between the origin O and the plane and m be the distance between the origin O and the point M(a, b, c). If a, b, c vary in the range of any positive numbers, then find the minimum value of $\left(\frac{m}{d}\right)^2$.

माना एक तल बिन्दु A(a, 0, 0), B(0, b, 0), C(0, 0, c) जहाँ a > 0, b > 0, c > 0, से गुजरता है। बिन्दु O के तल से दूरी d है तथा बिन्दु O तथा M(a, b, c) के मध्य दूरी m है। यदि a, b, c धनात्मक संख्याऐं हैं तो $\left(\frac{m}{d}\right)^2$ का न्यूनतम मान ज्ञात कीजिये।

Ans. 9

Sol. Plane $P = \frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$

$$d_{(0,0,0)} = d = \frac{1}{\sqrt{\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}}}$$

$$\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}=\frac{1}{d^2}$$

$m^2 = a^2 + b^2 + c^2$

Let $\varepsilon = \left(\frac{m}{d}\right)^2$

$$\varepsilon = \left(a^2+b^2+c^2\right)\left(\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}\right)$$

Find $\varepsilon_{min.}$

$$\varepsilon = 3+\frac{a^2}{b^2}+\frac{a^2}{c^2}+\frac{b^2}{a^2}+\frac{b^2}{c^2}+\frac{c^2}{a^2}+\frac{c^2}{b^2}$$

use AM & GM in last 6 terms

AM ≥ GM

$$\frac{\frac{a^2}{b^2}+\frac{a^2}{c^2}+\frac{b^2}{a^2}+\frac{b^2}{c^2}+\frac{c^2}{a^2}+\frac{c^2}{b^2}}{6} \geq (1)^{\frac{1}{6}}$$

$$\frac{a^2}{b^2}+\frac{a^2}{c^2}+\frac{b^2}{a^2}+\frac{b^2}{c^2}+\frac{c^2}{a^2}+\frac{c^2}{b^2} \geq 6$$

$\varepsilon_{min.} = 3 + 6 = 9$

6. Let A_1, A_2, A_3, A_4 be the areas of the triangular faces of a tetrahedron, and h_1, h_2, h_3, h_4 be the corresponding altitudes of the tetrahedron. If volume of tetrahedron is 5 cubic units, then find the minimum value of $(A_1 + A_2 + A_3 + A_4)(h_1 + h_2 + h_3 + h_4)$ (in cubic units).

माना A_1, A_2, A_3, A_4 एक चतुष्फलक के त्रिभुजाकार फलकों के क्षेत्रफल हैं एवं h_1, h_2, h_3, h_4 उनके संगत शीर्षलम्बों की लम्बाईयां हैं। यदि चतुष्फलक का आयतन 5 है तो $(A_1 + A_2 + A_3 + A_4)(h_1 + h_2 + h_3 + h_4)$ का न्यूनतम मान ज्ञात कीजिए।

Ans. 240

Sol. $\text{Volume}(V) = \frac{1}{3}A_1h_1$ or $h_1 = \frac{3V}{A_1}$

$h_2 = \frac{3V}{A_2}, h_3 = \frac{3V}{A_3}$ and $h_4 = \frac{3V}{A_4}$

$\left(A_1 + A_2 + A_3 + A_4\right)\left(h_1 + h_2 + h_3 + h_4\right)$

$= \left(A_1 + A_2 + A_3 + A_4\right)\left(\frac{3V}{A_1} + \frac{3V}{A_2} + \frac{3V}{A_3} + \frac{3V}{A_4}\right) = 3V\left(A_1 + A_2 + A_3 + A_4\right)\left(\frac{1}{A_1} + \frac{1}{A_2} + \frac{1}{A_3} + \frac{1}{A_4}\right)$

Now using A.M.-H.M inequality in A_1, A_2, A_3, A_4

$\frac{A_1 + A_2 + A_3 + A_4}{4} \geq \frac{4}{\left(\frac{1}{A_1} + \frac{1}{A_2} + \frac{1}{A_3} + \frac{1}{A_4}\right)}$

$\left(A_1 + A_2 + A_3 + A_4\right)\left(\frac{1}{A_1} + \frac{1}{A_2} + \frac{1}{A_3} + \frac{1}{A_4}\right) \geq 16$

Hence, the minimum value of $\left(A_1 + A_2 + A_3 + A_4\right)\left(h_1 + h_2 + h_3 + h_4\right) = 3V(16) = 48V = 48(5) = 240$

7. The vector $\vec{OP} = \hat{i} + 2\hat{j} + 2\hat{k}$ turns through a right angle, passing through the positive x-axis on the way. Find the vector in its new position.

यदि सदिश $\vec{OP} = \hat{i} + 2\hat{j} + 2\hat{k}$ को समकोण पर घुमाने के दौरान धनात्मक x-अक्ष को काटता है तो सदिश को नई स्थिति में ज्ञात कीजिये।

Ans. $\frac{4}{\sqrt{2}}\hat{i} - \frac{1}{\sqrt{2}}\hat{j} - \frac{1}{\sqrt{2}}\hat{k}$

Sol. $\overrightarrow{OP} = \hat{i} + 2\hat{j} + 2\hat{k}$

after rotation of $\overrightarrow{OP}$, let new vector is $\overrightarrow{OP}'$

Now $\overrightarrow{OP}$, $\hat{i}$, $\overrightarrow{OP}'$ will be coplanar

So $\overrightarrow{OP}' = \left|\overrightarrow{OP}\right| \dfrac{(\overrightarrow{OP} \times \hat{i}) \times \overrightarrow{OP}}{\left|(\overrightarrow{OP} \times \hat{i}) \times \overrightarrow{OP}\right|}$ $\left[\because \left|\overrightarrow{OP}\right| = \left|\overrightarrow{OP}'\right|\right]$

But $(\overrightarrow{OP} \times \hat{i}) \times \overrightarrow{OP} = 8\hat{i} - 2\hat{j} - 2\hat{k}$

$\Rightarrow \overrightarrow{OP}' = \dfrac{3(8\hat{i} - 2\hat{j} - 2\hat{k})}{2 \times 3\sqrt{2}}$

or $\overrightarrow{OP}' = \dfrac{4}{\sqrt{2}}\hat{i} - \dfrac{1}{\sqrt{2}}\hat{j} - \dfrac{1}{\sqrt{2}}\hat{k}$

8. If $\vec{a}$ and $\vec{b}$ are two vectors such that $|\vec{a}| = 1$, $|\vec{b}| = 4$, $\vec{a} \cdot \vec{b} = 2$. If $|\vec{a}| = 1$, $|\vec{b}| = 4$, $\vec{a} \cdot \vec{b} = 2$, $\vec{c} = (2\vec{a} \times \vec{b}) - 3\vec{b}$ then find the angle between $\vec{b}$ and $\vec{c}$.

यदि दो सदिश $\vec{a}$ तथा $\vec{b}$ इस प्रकार है कि $|\vec{a}| = 1$, $|\vec{b}| = 4$, $\vec{a} \cdot \vec{b} = 2$ यदि $\vec{c} = (2\vec{a} \times \vec{b}) - 3\vec{b}$ तो सदिश $\vec{b}$ एवं $\vec{c}$ के बीच का कोण बताइये।

Ans. $\dfrac{5\pi}{6}$

Sol. $|\vec{a}| = 1$

$|\vec{b}| = 4$

$|\vec{c}| = \left(2\vec{a} \times \vec{b}\right) - 3\vec{b}$

$\vec{a} \cdot \vec{b} = 2$

$\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos\theta$

$2 = 1 \times 4 \cos\theta$

$\cos\theta = \dfrac{1}{2} \Rightarrow \theta = 60°$

$|\vec{c}|^2 = \left|2\vec{a} \times \vec{b}\right|^2 + 9|\vec{b}|^2 - 2 \times 3\vec{a} \cdot \left(2\vec{a} \times \vec{b}\right)$

$|\vec{c}|^2 = 4\left|\vec{a} \times \vec{b}\right|^2 + 9|\vec{b}|^2$

$$|\vec{c}|^2 = 4\left|1\times 4\times\frac{\sqrt{3}}{2}\right|^2 + 9\times 16$$

$$|\vec{c}|^2 = 4\times 4\times 3 + 9\times 16$$

$$|\vec{c}|^2 = 16\times 12$$

$$|\vec{c}| = 8\sqrt{3}$$

$$\vec{b}\cdot\vec{c} = \vec{b}\cdot\left(2\vec{a}\times\vec{b}\right) - 3\vec{b}\cdot\vec{b}$$

$$4\times 8\sqrt{3}\cos\phi = 0 - 3\times 16$$

$$\cos\phi = -\frac{\sqrt{3}}{2}$$

$$\phi = \frac{5\pi}{6}$$

9. The pv's of the four angular points of a tetrahedron are: A $\left(\hat{j}+2\hat{k}\right)$; B $\left(3\hat{i}+\hat{k}\right)$; C $\left(4\hat{i}+3\hat{j}+6\hat{k}\right)$ and D $\left(2\hat{i}+3\hat{j}+2\hat{k}\right)$. Find :

(i) the perpendicular distance from A to the line BC.
(ii) the volume of the tetrahedron ABCD.
(iii) the perpendicular distance from D to the plane ABC.
(iv) the shortest distance between the lines AB and CD.

यदि चार बिन्दुओं की स्थिति सदिश क्रमशः A $\left(\hat{j}+2\hat{k}\right)$; B $\left(3\hat{i}+\hat{k}\right)$; C $\left(4\hat{i}+3\hat{j}+6\hat{k}\right)$ तथा D $\left(2\hat{i}+3\hat{j}+2\hat{k}\right)$ है तो Kkr d lhft , :

(i) । से रखा BC की लम्बवत् दूरी
(ii) चतुष्फलक ABCD का आयतन
(iii) क की तल ABC से लम्बवत् दूरी
(iv) । ठ एवं CD के मध्य न्यूनतम दूरी

Ans. (i) $\frac{6}{7}\sqrt{14}$ (ii) 6 (iii) $\frac{3}{5}\sqrt{10}$ (iv) $\sqrt{6}$

Sol. A(0,1,2)
B(3,0,1)
C(4,3,6)
D(2,3,2)

$\overrightarrow{AB}(3,-1,-1)$ $\overrightarrow{BC}(1,3,5)$ $\overrightarrow{CD}(-2,0,-4)$

(i) Line BC $\frac{x-3}{1}=\frac{y-0}{3}=\frac{z-1}{5}=\lambda$

Point on line BC P(3 + λ, 3λ, 1 + 5λ)

Let P is FOP from A to line BC

$\overrightarrow{AP}(3+\lambda,3\lambda,1+5\lambda)$

$\overrightarrow{AP}\cdot\overrightarrow{BC}=0$

$3 + \lambda + 9\lambda - 3 + 25\lambda - 5 = 0$

$35\lambda = 5 \Rightarrow \lambda = \frac{1}{7}$

$\overrightarrow{AP}\left(\frac{22}{7},-\frac{4}{7},-\frac{2}{7}\right)$

$\left|\overrightarrow{AP}\right| = \sqrt{\frac{484+16+4}{49}} = \sqrt{\frac{504}{49}} = \sqrt{\frac{36\times14}{49}} = \frac{6}{7}\sqrt{14}$

(ii) Volume of tetrahedron

$V = \frac{1}{6}[AB \quad AC \quad AD]$

$$\begin{vmatrix} 3 & -1 & -1 \\ 4 & 2 & 4 \\ 2 & 2 & 0 \end{vmatrix}$$

$= \frac{1}{6}\left|3(-8)+1(-8)-1(8-4)\right|$

$= \frac{1}{6}\left|24+8+4\right|$

$= \frac{1}{6}\times 36 = 6$

(iii) Plane ABC

$$\text{Normal to ABC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -1 & -1 \\ 1 & 3 & 5 \end{vmatrix}$$

$= \hat{i}(-5+3)-\hat{j}(15+1)+\hat{k}(10)$

$= -2\hat{i}-16\hat{j}+10\hat{k}$

<1, 8, –5>

Plane

$x + 8y - 5z = -2$

D(2,3,2)

$$d_{(2,3,2)} = \left|\frac{2+24-10}{\sqrt{1+64+25}}\right| = \frac{16}{\sqrt{90}} = \frac{16}{3\sqrt{10}} = \frac{8}{15}\sqrt{10}$$

(iv) Line AB $\frac{x}{3} = \frac{y-1}{-1} = \frac{z-2}{-1}$

Line CD $\frac{x-2}{-2} = \frac{y-3}{0} = \frac{z-2}{-4}$

$$\text{Shortest distance} = \overrightarrow{AC}\cdot\frac{\left(\overrightarrow{AB}\times\overrightarrow{CD}\right)}{\left|\overrightarrow{AB}\times\overrightarrow{CD}\right|}$$

$$\overrightarrow{AB}\times\overrightarrow{CD} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -1 & -1 \\ -2 & 0 & -4 \end{vmatrix}$$

$$\overrightarrow{AB}\times\overrightarrow{CD} = \left(+4\hat{i}+4\hat{j}-2\hat{k}\right)$$

$$= \hat{i}(+4) - \hat{j}(-12-2) + \hat{k}(-2)$$

$$\text{S.D.} = \left|(4,2,4)\cdot\frac{(+4,+14,-2)}{\sqrt{16+196+4}}\right|$$

$$= \left|\frac{16+28-8}{\sqrt{120}}\right|$$

$$= \left|\frac{36}{\sqrt{216}}\right|$$

$$= \left|\frac{36}{6\sqrt{6}}\right| = \sqrt{6}$$

10. If $\vec{a} = a_1\hat{i}+a_2\hat{j}+a_3\hat{k}$; $\vec{b} = b_1\hat{i}+b_2\hat{j}+b_3\hat{k}$ and $\vec{c} = c_1\hat{i}+c_2\hat{j}+c_3\hat{k}$ then show that the value of the scalar triple product $\left[n\vec{a}+\vec{b} \quad n\vec{b}+\vec{c} \quad n\vec{c}+\vec{a}\right]$ is $(n^3+1)\begin{vmatrix} \vec{a}\cdot\hat{i} & \vec{a}\cdot\hat{j} & \vec{a}\cdot\hat{k} \\ \vec{b}\cdot\hat{i} & \vec{b}\cdot\hat{j} & \vec{b}\cdot\hat{k} \\ \vec{c}\cdot\hat{i} & \vec{c}\cdot\hat{j} & \vec{c}\cdot\hat{k} \end{vmatrix}$.

यदि $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$; $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ तथा $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$ तो सिद्ध कीजिए कि $\left[n\vec{a} + \vec{b} \quad n\vec{b} + \vec{c} \quad n\vec{c} + \vec{a}\right]$

का मान $(n^3 + 1)$ $\begin{vmatrix} \vec{a}\cdot\hat{i} & \vec{a}\cdot\hat{j} & \vec{a}\cdot\hat{k} \\ \vec{b}\cdot\hat{i} & \vec{b}\cdot\hat{j} & \vec{b}\cdot\hat{k} \\ \vec{c}\cdot\hat{i} & \vec{c}\cdot\hat{j} & \vec{c}\cdot\hat{k} \end{vmatrix}$ होगा।

Ans.

Sol. $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$

$\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$

$\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$

$[\vec{a}\,\vec{b}\,\vec{c}]$ is written as $\begin{vmatrix} \vec{a}.\hat{i} & \vec{a}.\hat{j} & \vec{a}.\hat{k} \\ \vec{b}.\hat{i} & \vec{b}.\hat{j} & \vec{b}.\hat{k} \\ \vec{c}.\hat{i} & \vec{c}.\hat{j} & \vec{c}.\hat{k} \end{vmatrix}$

Now $\{(n\vec{a} + \vec{b}) \times (n\vec{b} + \vec{c})\}.(n\vec{c} + \vec{a})$

$= \{n^2(\vec{a} \times \vec{b}) + n(\vec{a} \times \vec{c}) + \vec{b} \times \vec{c}\} \cdot (n\vec{c} + \vec{a})$

$= n^3[\vec{a}\,\vec{b}\,\vec{c}] + [\vec{b}\,\vec{c}\,\vec{a}]$

$= (n^3 + 1)\ [\vec{a}\,\vec{b}\,\vec{c}]$

11. ABCD is a tetrahedron with pv's of its angular points as A(–5, 22, 5); B(1, 2, 3); C(4, 3, 2) and D(–1, 2, – 3). If the area of the triangle AEF where the quadrilaterals ABDE and ABCF are parallelograms is $\sqrt{S}$ then find the value of S.

। ठ ब क एक चतुष्फलक है तथा इनके शीर्ष की स्थिति सदिश क्रमशः A(–5, 22, 5); B(1, 2, 3); C(4, 3, 2) एवं D(–1, 2, – 3) है। यदि ABDF एवं ABCF एक समान्तर चतुर्भुज है तथा त्रिभुज AEF का क्षेत्रफल $\sqrt{S}$ हो तो S का मान बताइये।

Ans. 110

Sol. A(–5,22,5)
B(1,2,3)
C(4,3,2)
D(–1,2,–3)
ABDE is parallelogram
Let E(α, β, γ)

Mid–point of AD = Mid–point of BE

$$(-3,12,1)=\left(\frac{\alpha+1}{2},\frac{\beta+2}{2},\frac{\gamma+3}{2}\right)$$

$\alpha = -7$

$\beta = 22$

$\gamma = -1$

E(–7, 22, –1)

$\overrightarrow{AE}=(-2,0,-6)$

ABCF is a parallelogram

Let F(a,b,c)

Mid–point of AC = mid–point of BF

$$\left(-\frac{1}{2},\frac{25}{2},\frac{7}{2}\right)=\left(\frac{1+a}{2},\frac{2+b}{2},\frac{3+c}{2}\right)$$

a = –2

b = 23

c = 4

F(–2,23,4)

$\overrightarrow{AF}=(3,1,-1)$

Area of ΔAEF

$$=\frac{1}{2}\left|\overrightarrow{AE}\times\overrightarrow{AF}\right|$$

$$=\frac{1}{2}\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\-2&0&-6\\3&1&-1\end{vmatrix}$$

$$=\frac{1}{2}\left|\hat{i}(6)-\hat{j}(2+18)+\hat{k}(-2)\right|$$

$$=\frac{1}{2}\sqrt{36+400+4}$$

$$=\frac{1}{2}\sqrt{440}$$

$$=\sqrt{110}$$

12. If volume of a regular tetrahedron of edge length k is V and shortest distance between any pair of opposite edges of same regular tetrahedron is d, then find the value of $\frac{d^3}{V}$.

एक समान्तर चतुष्फलक जिसके भुजा की लम्बाई k है का आयतन V है। यदि किन्हीं दो विपरीत भुजाओं के मध्य न्यूनतम दूरी d है तो $\frac{d^3}{V}$ का मान ज्ञात कीजिये।

Ans. 3

Sol. $V = \frac{1}{6}\left|\left[\vec{a}\ \vec{b}\ \vec{c}\right]\right|$ = volume of tetrahedron OABC

$$\text{Noω, } \left[\vec{a}\ \vec{b}\ \vec{c}\right]^2 = \begin{vmatrix} \vec{a}\cdot\vec{a} & \vec{a}\cdot\vec{b} & \vec{a}\cdot\vec{c} \\ \vec{b}\cdot\vec{a} & \vec{b}\cdot\vec{b} & \vec{b}\cdot\vec{c} \\ \vec{c}\cdot\vec{a} & \vec{c}\cdot\vec{b} & \vec{c}\cdot\vec{c} \end{vmatrix} = k^6 \begin{vmatrix} 1 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 1 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 1 \end{vmatrix} = \frac{k^6}{2} \Rightarrow \left|\left[\vec{a}\ \vec{b}\ \vec{c}\right]\right| = \frac{k^3}{\sqrt{2}}$$

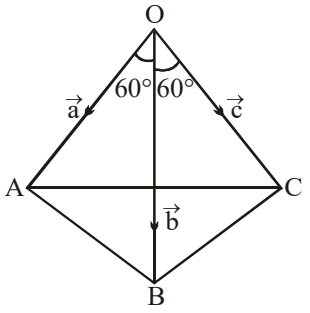
So, $V = \frac{k^3}{6\sqrt{2}}$

Shortest distance between OA and BC, is

$$d = \left|\frac{\left(\vec{b}-\vec{a}\right)\cdot\left(\vec{a}\times\left(\vec{c}-\vec{b}\right)\right)}{\left|\vec{a}\times\left(\vec{c}-\vec{b}\right)\right|}\right| = \left|\frac{\left[\vec{a}\ \vec{c}\ \vec{b}\right]}{\left|\vec{a}\times\left(\vec{c}-\vec{b}\right)\right|}\right| = \frac{k^3/\sqrt{2}}{k^2} = \frac{k}{\sqrt{2}}, \text{ as}$$

$$\left|\vec{a}\times\left(\vec{c}-\vec{b}\right)\right|^2 = \vec{a}^2(\vec{c}^2-\vec{b}^2) - \left(\vec{a}\cdot\left(\vec{c}-\vec{b}\right)\right)^2 = \vec{a}^2(\vec{c}^2+\vec{b}^2-2\vec{c}\cdot\vec{b}) - \left(\vec{a}\cdot\left(\vec{c}-\vec{b}\right)\right)^2 = k^4 + k^4 - \frac{2k^2k^2}{2} = k^4 \Rightarrow$$

$\left|\vec{a}\times\left(\vec{c}-\vec{b}\right)\right| = k^2$.



Hence $\frac{d^3}{V} = \frac{\left(\frac{k}{\sqrt{2}}\right)^3}{\frac{1}{6\sqrt{2}}k^3} = \frac{1}{2\sqrt{2}}\left(6\sqrt{2}\right) = 3$. **Ans.**

13. Find the scalars α and β if $\vec{a}\times(\vec{b}\times\vec{c}) + (\vec{a}.\vec{b})\vec{b} = (4-2\beta-\sin\alpha)\vec{b} + (\beta^2-1)\vec{c}$ and $(\vec{c}.\vec{c})\vec{a} = \vec{c}$ while $\vec{b}$ and $\vec{c}$ are non zero non collinear vectors.

दिया है $\vec{a}\times(\vec{b}\times\vec{c}) + (\vec{a}.\vec{b})\vec{b} = (4-2\beta-\sin\alpha)\vec{b} + (\beta^2-1)\vec{c}$ एवं $(\vec{c}.\vec{c})\vec{a} = \vec{c}$ यदि $\vec{b}$ एवं $\vec{c}$ अशून्य तथा असंरेखीय है तो α एवं β के मान ज्ञात कीजिए।

Ans. $\alpha = n\pi + \frac{(-1)^n\pi}{2}, n \in I \& \beta = 1$

Sol. $\vec{a}\times\left(\vec{b}\times\vec{c}\right)+\left(\vec{a}\cdot\vec{b}\right)\vec{b}=\left(4-2\beta-\sin\alpha\right)\vec{b}+\left(\beta^2-1\right)\vec{c}$

$\left(\vec{c}\cdot\vec{c}\right)\vec{a}=\vec{c}$

$\vec{a}\times\left(\vec{b}\times\vec{c}\right)=\vec{b}\left(\vec{a}\cdot\vec{c}\right)-\vec{c}\left(\vec{a}\cdot\vec{b}\right)$

$\vec{a}\times\left(\vec{b}\times\vec{c}\right)+\left(\vec{a}\cdot\vec{b}\right)\vec{b}=\vec{b}\left(\vec{a}\cdot\vec{c}+\vec{a}\cdot\vec{b}\right)-\vec{c}\left(\vec{a}\cdot\vec{b}\right)$

By comparing with given relation

$\vec{a}\cdot\vec{c}+\vec{a}\cdot\vec{b}=4-2\beta-\sin\alpha$ ...(1)

$-\vec{a}\cdot\vec{b}=\beta^2-1$ ...(2)

Now $\left(\vec{c}\cdot\vec{c}\right)\vec{a}=\vec{c}$

$\left(\vec{c}\cdot\vec{c}\right)\vec{a}\cdot\vec{c}=\vec{c}\cdot\vec{c}$

$\vec{a}\cdot\vec{c}=1$ ...(3)

Using eq (1), (2) & (3)

$1=\beta^2-1+4-2\beta-\sin\alpha$

$\beta^2-2\alpha+2=\sin\alpha$

$(\beta-1)^2+1=\sin\alpha$

$\sin\alpha=1$ $\beta=1$

$\alpha=n\pi+(-1)^n\frac{\pi}{2}$ $\beta=1$

14. Let $\vec{a}=\alpha\hat{i}+2\hat{j}-3\hat{k}$, $\vec{b}=\hat{i}+2\alpha\hat{j}-2\hat{k}$ and $\vec{c}=2\hat{i}-\alpha\hat{j}+\hat{k}$. Find the value(s) of α, if any, such that $\left\{\left(\vec{a}\times\vec{b}\right)\times\left(\vec{b}\times\vec{c}\right)\right\}\times\left(\vec{c}\times\vec{a}\right)=0$. Find the vector product when $\alpha=0$.

यदि $\vec{a}=\alpha\hat{i}+2\hat{j}-3\hat{k}$, $\vec{b}=\hat{i}+2\alpha\hat{j}-2\hat{k}$ तथा $\vec{c}=2\hat{i}-\alpha\hat{j}+\hat{k}$ यदि $\left\{\left(\vec{a}\times\vec{b}\right)\times\left(\vec{b}\times\vec{c}\right)\right\}\times\left(\vec{c}\times\vec{a}\right)=0$ है तो α का मान ज्ञात करो। यदि $\alpha=0$ तो उनका सदिश गुणनफल बताओ।

Ans. $\alpha=2/3$; if $\alpha=0$ then vector product is $-60\left(2\hat{i}+\hat{k}\right)$

Sol. $\vec{a}=\langle\alpha, 2, -3\rangle$

$\vec{b}=\langle1, 2\alpha, -2\rangle$

$\vec{c}=\langle2, -\alpha, 1\rangle$

Case-I

$\left(\vec{a}\times\vec{b}\right)\times\left(\vec{b}\times\vec{c}\right)=\lambda\left(\vec{c}\times\vec{a}\right)$

dot with b

$\vec{b}\left(\left(\vec{a}\times\vec{b}\right)\times\left(\vec{b}\times\vec{c}\right)\right)=\lambda\left[\vec{a}\quad\vec{b}\quad\vec{c}\right]$

$\left(\vec{b}\times\left(\vec{a}\times\vec{b}\right)\cdot\left(\vec{b}\times\vec{c}\right)\right)=\lambda\left[\vec{a}\quad\vec{b}\quad\vec{c}\right]$

$\left(\vec{a}\left(\vec{b}\cdot\vec{b}\right)-\vec{b}\left(\vec{a}\cdot\vec{b}\right)\right)\cdot\left(\vec{b}\times\vec{c}\right)=\lambda\left[\vec{a}\quad\vec{b}\quad\vec{c}\right]$

$\left(\vec{b}\cdot\vec{b}\right)\left[\vec{a}\quad\vec{b}\quad\vec{c}\right]-\left[\vec{a}\quad\vec{b}\right]\times 0=\lambda\left[\vec{a}\quad\vec{b}\quad\vec{c}\right]$

$\vec{b}\cdot\vec{b}\left[\vec{a}\quad\vec{b}\quad\vec{c}\right]=\lambda\left[\vec{a}\quad\vec{b}\quad\vec{c}\right]$

$\left[\vec{a}\quad\vec{b}\quad\vec{c}\right]=0$

$$\left[\vec{a}\quad\vec{b}\quad\vec{c}\right]=\begin{vmatrix}\alpha & 2 & -3\\ 1 & 2\alpha & -2\\ 2 & -\alpha & 1\end{vmatrix}=0$$

$= \alpha(2\alpha - 2\alpha) - 2(1 + 4) - 3\,(-\alpha - 4\alpha) = 0$

$0 - 10 + 15\alpha = 0$

$\alpha=\frac{2}{3}$

Case-II : $\vec{c}\times\vec{a}=0$ Not possible

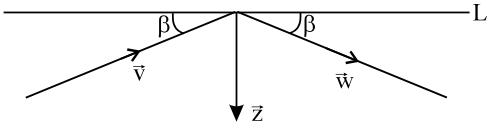
Case-III : $\left(\vec{a}\times\vec{b}\right)=\lambda\left(\vec{b}\times\vec{c}\right)$

dot with $\vec{c}$

$\left[\vec{a}\quad\vec{b}\quad\vec{c}\right]=0$

15. The figure shows non zero vector $\vec{v}$, $\vec{w}$ and $\vec{z}$ with orthogonal to the line L, and $\vec{v}$ and $\vec{w}$ making equal angles β with the line L. Assuming $|\vec{v}|=|\vec{w}|$, if the vector $\vec{w}$ is expressed as a linear combination of $\vec{v}$ and $\vec{z}$ as $\vec{w}=x\vec{v}+y\vec{z}$. Find the value of x and y.

दिखाए गये चित्रनुसार तीन अशून्य सदिश $\vec{v}$, $\vec{w}$ एवं $\vec{z}$ है जहाँ $\vec{z}$ रेखा L के लम्बवत् (orthogonal) है तथा $\vec{v}$ एवं $\vec{w}$ रेखा L के साथ l eku d kskβ बनाती है। $|\vec{v}|=|\vec{w}|$, $\vec{w}$ को $\vec{v}$ तथा $\vec{z}$ के रेखिक संचय के रूप में व्यक्त किया जा सके अर्थात् $\vec{w}=x\vec{v}+y\vec{z}$ तो x एवं y का मान ज्ञात कीजिए।



Ans. $x = 1$ and $y=\frac{2|\vec{v}|\sin\beta}{|\vec{z}|}$

Sol. $\vec{w}=x\vec{v}+y\vec{z}$ $|\vec{v}|=|\vec{w}|$

dot $\vec{v}$

$\vec{w}\cdot\vec{v} = x|\vec{v}|^2 + yz\cdot\vec{v}$

$|\vec{v}||\vec{w}|\cos 2\beta = x|\vec{v}|^2 + y|\vec{z}||\vec{v}|\cos(90+\beta)$

$|\vec{w}|\cos 2\beta = x|\vec{w}| - y|\vec{z}|\sin\beta$ ...(1)

$\vec{w} = x\vec{v} + y\vec{z}$

dot with $\vec{w}$

$|\vec{w}|^2 = x|\vec{v}||\vec{w}|\cos 2\beta + y|\vec{z}||\vec{w}|\cos(90-\beta)$

$|\vec{w}| = x|\vec{w}|\cos 2\beta + y|\vec{z}|\sin\beta$ ...(2)

(1) + (2)

$|\vec{w}|(1+\cos 2\beta) = x(1+\cos 2\beta)|\vec{w}|$

$x = 1$

$y = \dfrac{|\vec{w}|(1-\cos 2\beta)}{|\vec{z}|\sin\beta}$

$= \dfrac{|\vec{w}|2\sin\beta}{|\vec{z}|}$

16. Consider the non zero vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ and $\vec{d}$ such that no three of which are coplanar then prove that $\vec{a}\left[\vec{b}\,\vec{c}\,\vec{d}\right] + \vec{c}\left[\vec{a}\,\vec{b}\,\vec{d}\right] = \vec{b}\left[\vec{a}\,\vec{c}\,\vec{d}\right] + \vec{d}\left[\vec{a}\,\vec{b}\,\vec{c}\right]$. Hence prove that if $\vec{a}$, $\vec{b}$, $\vec{c}$ and $\vec{d}$ represent the position vectors of the vertices of a plane quadrilateral then $\dfrac{\left[\vec{b}\,\vec{c}\,\vec{d}\right] + \left[\vec{a}\,\vec{b}\,\vec{d}\right]}{\left[\vec{a}\,\vec{c}\,\vec{d}\right] + \left[\vec{a}\,\vec{b}\,\vec{c}\right]} = 1$.

यदि चार अशून्य सदिश $\vec{a}$, $\vec{b}$, $\vec{c}$ तथा $\vec{d}$ इस प्रकार से है कि उनमें से कोई भी तीन समतलीय नही हो तो सिद्ध करो कि $\vec{a}\left[\vec{b}\,\vec{c}\,\vec{d}\right] + \vec{c}\left[\vec{a}\,\vec{b}\,\vec{d}\right] = \vec{b}\left[\vec{a}\,\vec{c}\,\vec{d}\right] + \vec{d}\left[\vec{a}\,\vec{b}\,\vec{c}\right]$. $\vec{a}$, $\vec{b}$, $\vec{c}$ तथा $\vec{d}$ यदि एक चतुर्भुज के शीर्षो की स्थिति सदिश है तो सिद्ध d lfr , fd $\dfrac{\left[\vec{b}\,\vec{c}\,\vec{d}\right] + \left[\vec{a}\,\vec{b}\,\vec{d}\right]}{\left[\vec{a}\,\vec{c}\,\vec{d}\right] + \left[\vec{a}\,\vec{b}\,\vec{c}\right]} = 1$

Ans.

Sol. (1) $\vec{a} = x\vec{b} + y\vec{c} + z\vec{d}$

$\left[\vec{a}\quad\vec{b}\quad\vec{c}\right] = z\left[\vec{d}\quad\vec{b}\quad\vec{c}\right] = z\left[\vec{b}\quad\vec{c}\quad\vec{d}\right]$

$\left[\vec{a}\ \ \vec{c}\ \ \vec{d}\right] = x\left[\vec{b}\ \ \vec{c}\ \ \vec{d}\right]$

$\left[\vec{a}\ \ \vec{b}\ \ \vec{d}\right] = y\left[\vec{c}\ \ \vec{b}\ \ \vec{d}\right] = -y\left[\vec{b}\ \ \vec{c}\ \ \vec{d}\right]$

LHS $\vec{a}\left[\vec{b}\ \ \vec{c}\ \ \vec{d}\right] = \vec{c}\left[\vec{a}\ \ \vec{b}\ \ \vec{d}\right]$

$\vec{a}\left[\vec{b}\ \ \vec{c}\ \ \vec{d}\right] = -y\vec{c}\left[\vec{b}\ \ +\vec{c}\ \ \vec{d}\right]$

RHS $\vec{b}\left[\vec{a}\ \ \vec{c}\ \ \vec{d}\right] = \vec{b}x\left[\vec{b}\ \ \vec{c}\ \ \vec{d}\right]$

$+\vec{d}\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right] = +z\vec{d}\left[\vec{b}\ \ +\vec{c}\ \ +\vec{d}\right]$

putting in equation

$\left(\vec{b}\ \ \vec{c}\ \ \vec{d}\right)\left(\vec{a}\ \ -y\vec{c}\right) = \left[\vec{b}\ \ \vec{c}\ \ \vec{d}\right]\left[x\vec{b}\ \ +z\vec{d}\right]$

$\vec{a} = x\vec{b} + y\vec{c} + z\vec{d}$

(2) $\vec{a}\ \ \vec{b}\ \ \vec{c}\ \ \vec{d}$ $\Rightarrow$ PV of vertices of a plane quadrilateral

$\left[\left(\vec{a}-\vec{b}\right)\times\left(\vec{b}-\vec{c}\right)\right].\left(\vec{c}-\vec{d}\right) = 0$

$\left(\vec{a}\times\vec{b} - \vec{a}\times\vec{c} + \vec{b}\times\vec{c}\right)\cdot\left(\vec{c}-\vec{d}\right) = 0$

$\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right] - \left[\vec{a}\ \ \vec{b}\ \ \vec{d}\right] + \left[\vec{a}\ \ \vec{c}\ \ \vec{d}\right] - \left[\vec{b}\ \ \vec{c}\ \ \vec{d}\right] = 0$

$$\frac{\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right] + \left[\vec{a}\ \ \vec{c}\ \ \vec{d}\right]}{\left[\vec{a}\ \ \vec{b}\ \ \vec{d}\right] + \left[\vec{b}\ \ \vec{c}\ \ \vec{d}\right]} = 1$$

17. Let $\vec{A} = \hat{i} - 2\hat{j} + 3\hat{k}$, $\vec{B} = 2\hat{i} + \hat{j} - \hat{k}$, $\vec{C} = \hat{j} + \hat{k}$. If the vector $\vec{B}\times\vec{C}$ can be expressed as a linear combination $\vec{B}\times\vec{C} = x\vec{A} + y\vec{B} + z\vec{C}$ where x, y, z are scalars, then find the value of (100x + 10y + 8z).

माना $\vec{A} = \hat{i} - 2\hat{j} + 3\hat{k}$, $\vec{B} = 2\hat{i} + \hat{j} - \hat{k}$, $\vec{C} = \hat{j} + \hat{k}$. यदि सदिश $\vec{B}\times\vec{C}$ को रैखिक संचय के रूप में $\vec{B}\times\vec{C} = x\vec{A} + y\vec{B} + z\vec{C}$ तरह से व्यक्त किया जा सके जहाँ x, y, z अदिश है तो (100x + 10y + 8z) का मान ज्ञात कीजिये।

Ans. 101

Sol. $\vec{A}(1,-2,3)$

$\vec{B}(2,1,-1)$

$\vec{C}(0,1,1)$

$\vec{B}\times\vec{C} = x\vec{A} + y\vec{B} + z\vec{C}$

$$\vec{B}\times\vec{C}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 2 & 1 & -1\\ 0 & 1 & 1\end{vmatrix}=2(1,-1,1)$$

$2(1,-1,1) = x(1,-2,3) + y(2,1,-1) + z(0,1,1)$

$x + 2y = 2$

$-2x + y + z = -2$

$3x - y + z = 2$

Solving

$$x = 1,\ y = \frac{1}{2},\ z = -\frac{1}{2}$$

$100x + 10y + 8z = 100 + 5 - 4 = 101$

18. Let $\vec{a}=\begin{bmatrix}1\\0\\-3\end{bmatrix}$; $\vec{b}=\begin{bmatrix}2\\1\\0\end{bmatrix}$; $\vec{c}=\begin{bmatrix}1\\-1\\1\end{bmatrix}$. Find the numbers α, β, γ such that $\alpha\vec{a}+\beta\vec{b}+\gamma\vec{c}=\begin{bmatrix}-2\\-5\\6\end{bmatrix}$.

दिया है $\vec{a}=\begin{bmatrix}1\\0\\-3\end{bmatrix}$; $\vec{b}=\begin{bmatrix}2\\1\\0\end{bmatrix}$; $\vec{c}=\begin{bmatrix}1\\-1\\1\end{bmatrix}$. यदि α, β, γ तीन अदिश ऐसे है कि $\alpha\vec{a}+\beta\vec{b}+\gamma\vec{c}=\begin{bmatrix}-2\\-5\\6\end{bmatrix}$ तो α, β, γ ज्ञात कीजिए।

Ans. $\alpha = -1, \beta = -2, \gamma = 3$

Sol. $\vec{a}=\begin{bmatrix}1\\0\\-3\end{bmatrix}$ $\vec{b}=\begin{bmatrix}2\\1\\0\end{bmatrix}$ $\vec{c}=\begin{bmatrix}1\\-1\\1\end{bmatrix}$

$$\alpha\vec{a}+\beta\vec{b}+\gamma\vec{c}=\begin{bmatrix}-2\\-5\\6\end{bmatrix}$$

$\alpha + 2\beta + \gamma = -2$

$\beta - \gamma = -5$

$-3\alpha + \gamma = 6$

Solving them

$\alpha = -1$

$\beta = -2$

$\gamma = 3$

19. Let a 3 dimensional vector $\vec{V}$ satisfies the condition, $2\vec{V} + \vec{V} \times (\hat{i} + 2\hat{j}) = 2\hat{i} + \hat{k}$. If $3|\vec{V}| = \sqrt{m}$ where $m \in N$, then find m.

माना एक त्रिविम सदिश $\vec{V}$ प्रतिबन्ध $2\vec{V} + \vec{V} \times (\hat{i} + 2\hat{j}) = 2\hat{i} + \hat{k}$ को सन्तुष्ट करता है। यदि $3|\vec{V}| = \sqrt{m}$ जहाँ $m \in N$ तो m ज्ञात कीजिये।

Ans. 6

Sol. $2\vec{V} + \vec{V} \times \left(\hat{i} + 2\hat{j}\right) = 2\hat{i} + \hat{k}$

$3|\vec{V}| = \sqrt{m}$

dot with $\left(\hat{i} + 2\hat{j}\right)$

$2\vec{V} \cdot \left(\hat{i} + 2\hat{j}\right) = 2$

$\vec{V} \cdot \left(\hat{i} + 2\hat{j}\right) = 1$

$\left|2\vec{V} + \vec{V} \times \left(\hat{i} + 2\hat{j}\right)\right|^2 = \left(2\hat{j} + \hat{k}\right)^2$

$4|\vec{V}|^2 + \left(\vec{V} \times \left(\hat{i} + 2\hat{j}\right)\right) \cdot \left(\vec{V} \times \left(\hat{i} + 2\hat{j}\right)\right) + 2\vec{V} \cdot \left(\vec{V} \times \left(\hat{i} + 2\hat{j}\right)\right) = \left(\sqrt{4+1}\right)^2$

$4|\vec{V}|^2 + \vec{V} \cdot \left(\left(\hat{i} + 2\hat{j}\right) \times \left(\vec{V}\left(\hat{i} + 2\hat{j}\right)\right)\right) = 5$

$4|\vec{V}|^2 + \vec{V} \cdot \left(\vec{V}(1+4) - \left(\hat{i} + 2\hat{j}\right)\left(\vec{V} \cdot \left(\hat{i} + 2\hat{j}\right)\right)\right)$

$4|\vec{V}|^2 + \vec{V} \cdot \left(5\vec{V} - \left(\hat{i} + 2\hat{j}\right)(1)\right) = 5$

$9|\vec{V}|^2 - \vec{V} \cdot \left(\hat{i} + 2\hat{j}\right) = 5$

$9|\vec{V}|^2 = 6$

$3|\vec{V}| = \sqrt{6}$

20. If $\vec{x}$, $\vec{y}$ are two non-zero and non-collinear vectors satisfying

$[(a-2)\alpha^2 + (b-3)\alpha + c]\vec{x} + [(a-2)\beta^2 + (b-3)\beta + c]\vec{y} + [(a-2)\gamma^2 + (b-3)\gamma + c](\vec{x} \times \vec{y}) = 0$
where α, β, γ are three distinct real numbers, then find the value of $(a^2 + b^2 + c^2)$.

यदि $\vec{x}$, $\vec{y}$ दो अशून्य विषम रेखिये सदिश हैं जो

$[(a-2)\alpha^2 + (b-3)\alpha + c]\vec{x} + [(a-2)\beta^2 + (b-3)\beta + c]\vec{y} + [(a-2)\gamma^2 + (b-3)\gamma + c](\vec{x} \times \vec{y}) = 0$
को सन्तुष्ट करते हैं जहाँ α, β, γ तीन भिन्न वास्तविक संख्याऐं हैं तो $(a^2 + b^2 + c^2)$ का मान ज्ञात कीजिये।

Ans. 13

Sol. Clearly, $(a-2)\alpha^2 + (b-3)\alpha + c = 0$

$(a-2)\beta^2 + (b-3)\beta + c = 0$ } (As $\vec{x}, \vec{y}$ and $(\vec{x} \times \vec{y})$ are non coplanar.) and

$(a-2)\gamma^2 + (b-3)\gamma + c = 0$

$\therefore$ The equation $(a-2)x^2 + (b-3)x + c = 0$ has 3 distinct roots.

$\therefore$ Above equation will be an identity

$\therefore\ a - 2 = 0 \Rightarrow a = 2$

$b - 3 = 0 \Rightarrow b = 3$

$c = 0$

$\therefore\ a^2 + b^2 + c^2 = 13$ Ans.

21. The diagonals of a parallelogram are given by vectors $2\hat{i} + 3\hat{j} - 6\hat{k}$ and $3\hat{i} - 4\hat{j} - \hat{k}$. Determine its sides and also the area.

किसी समान्तर चतुर्भुज के विकर्णो के सदिश क्रमशः $2\hat{i} + 3\hat{j} - 6\hat{k}$ तथा $3\hat{i} - 4\hat{j} - \hat{k}$ है। उसकी भुजाओं तथा क्षेत्रफल ज्ञात कीजिए।

Ans. $\frac{1}{2}\left(5\hat{i} - \hat{j} - 7\hat{k}\right), \frac{1}{2}(-\hat{i} + 7\hat{j} - 5\hat{k});\ \frac{1}{2}\sqrt{1274}$ sq. units

Sol. $\vec{a} + \vec{b} = (2,3,-6) = \vec{d}_1$

$\vec{a} - \vec{b} = (3,4,-1) = \vec{d}_2$

$\vec{a} = \left(\frac{5}{2}, -\frac{1}{2}, -\frac{7}{2}\right)$

$\vec{b} = \left(-\frac{1}{2}, +\frac{7}{2}, -\frac{5}{2}\right)$

Area $= \vec{d}_1 \times \vec{d}_2$

$$= \frac{1}{2}\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & -6 \\ 3 & -4 & -1 \end{vmatrix}$$

$$= \frac{1}{2}\left|\hat{i}(-3-24) - \hat{j}(-2+18) + \hat{k}(-8-9)\right|$$

$$= \frac{1}{2}\left|-27\hat{i} - 16\hat{j} - 17\hat{k}\right|$$

$$= \frac{1}{2}\sqrt{1274}$$

22. Find the value of λ such that a, b, c are all non-zero and

$\left(-4\hat{i}+5\hat{j}\right)a+(3\hat{i}-3\hat{j}+\hat{k})b+(\hat{i}+\hat{j}+3\hat{k})c=\lambda(a\hat{i}+b\hat{j}+c\hat{k})$

a, b, c अशून्य संख्याओं के लिये λ का मान ज्ञात कीजिये। कि $\left(-4\hat{i}+5\hat{j}\right)a+(3\hat{i}-3\hat{j}+\hat{k})b+(\hat{i}+\hat{j}+3\hat{k})c=\lambda(a\hat{i}+b\hat{j}+c\hat{k})$ है।

Ans. $\lambda = 0,\ \lambda = -2 \pm \sqrt{29}$

Sol. $-4a + 3b + c = a\lambda$

$(-4 - \lambda)\, a + 3b + c = 0$ ...(i)

$5a - 3b + xc = \lambda b$

$5a - (3 + \lambda)b + c = 0$ ...(ii)

$b + 3c = \lambda c$

$b + (3 - \lambda)c = 0$ ...(iii)

Homogeneous system with non–zero solution

$$\begin{vmatrix} -4-\lambda & 3 & 1 \\ 5 & -3-\lambda & 1 \\ 0 & 1 & 3-\lambda \end{vmatrix} = 0$$

$(-4 - \lambda)((-3 - \lambda)(3 - \lambda) - 1) - 5(9 - \lambda - 1) = 0$

$(4 + \lambda)(9 - \lambda^2 + 1) - 5(8 - \lambda) = 0$

$40 - 4\lambda^2 - 10\lambda - \lambda^3 - 40 + 5\lambda = 0$

$\lambda^3 + 4\lambda^2 + 15\lambda = 0$

$d = 0 \qquad \lambda^2 + 4\lambda - 15 = 0$

$$\lambda = \frac{-4 \pm \sqrt{16+60}}{4}$$

$\lambda = -4 \pm \sqrt{29}$

23. Find the vector $\vec{r}$ which is perpendicular to $\vec{a} = \hat{i} - 2\hat{j} + 5\hat{k}$ and $\vec{b} = 2\hat{i} + 3\hat{j} - \hat{k}$ and $\vec{r} \cdot \left(2\hat{i} + \hat{j} + \hat{k}\right) + 8 = 0$.

$\vec{a} = \hat{i} - 2\hat{j} + 5\hat{k}$ तथा $\vec{b} = 2\hat{i} + 3\hat{j} - \hat{k}$ के लम्बवत् सदिश $\vec{r}$ इस प्रकार है कि $\vec{r} \cdot \left(2\hat{i} + \hat{j} + \hat{k}\right) + 8 = 0$

Ans. $\vec{r} = -13\hat{i} + 11\hat{j} + 7\hat{k}$

Sol. $\vec{r} = \lambda\left(\vec{a} \times \vec{b}\right)$

$$= \lambda \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & 5 \\ 2 & 3 & -1 \end{vmatrix}$$

$= \lambda\left(-13\hat{i} + 11\hat{j} + 7\hat{k}\right)$

$r \cdot (2,1,1) = -8$
$\lambda(-26 + 11 + 7) = -8$
$\lambda(-8) = -8$
$\lambda = 1$

24. Two vertices of a triangle are at $-\hat{i}+3\hat{j}$ and $2\hat{i}+5\hat{j}$ and its orthocentre is at $\hat{i}+2\hat{j}$. Find the position vector of third vertex.

किसी त्रिभुज के दो शीर्षो $-\hat{i}+3\hat{j}$ तथा $2\hat{i}+5\hat{j}$ और लम्बकेन्द्र $\hat{i}+2\hat{j}$ है। तो तीसरे शीर्ष का स्थिति सदिश ज्ञात कीजिए।

Ans. $\frac{5}{7}\hat{i}+\frac{17}{7}\hat{j}$

Sol. Orthocenter O (1,2)
DR′s of BE = (1, 3)
DR′s of BF = (2, –1)
DR′s of AC = $(\alpha + 1, \beta - 3)$
DR′s of BC = $((\alpha - 2), (\beta - 5)$
BE $\perp$ AC
$(1, 3) \cdot (\alpha + 1, \beta - 3) = 0$
$\alpha + 1 + 3\beta - 9 = 0$
$\alpha + 3\beta = 8$ ...(i)
BF $\perp$ BC
$(2, -1) \cdot (\alpha - 2, \beta - 5) = 0$
$2\alpha - 4 - \beta + 5 = 0$
$2\alpha - \beta = -1$ ...(ii)
equation (i) + 3× (ii)
$7\alpha = 5$

$$\alpha = \frac{5}{7}$$

$$\beta = \frac{10}{7} + 1 = \frac{17}{7}$$

Third vertex $\left(\frac{5}{7}, \frac{17}{7}\right)$

25. Let $\vec{A}(t) = f_1(t)\hat{i} + f_2(t)\hat{j}$ and $\vec{B}(t) = g_1(t)\hat{i} + g_2(t)\hat{j}$, $t \in [0, 1]$, where f_1, f_2, g_1, g_2 are continuous functions. If $\vec{A}(t)$ and $\vec{B}(t)$ are nonzero vectors for all t and $\vec{A}(0) = 2\hat{i}+3\hat{j}$, $\vec{A}(1) = 6\hat{i}+2\hat{j}$, $\vec{B}(0) = 3\hat{i}+2\hat{j}$ and $\vec{B}(1) = 2\hat{i}+6\hat{j}$, then show that $\vec{A}(t)$ and $\vec{B}(t)$ are parallel for some t.

माना कि $\vec{A}(t) = f_1(t)\hat{i} + f_2(t)\hat{j}$ तथा $\vec{B}(t) = g_1(t)\hat{i} + g_2(t)\hat{j}$, $t \in [0, 1]$, है जहाँ f_1, f_2, g_1, g_2 सतत् फलन है यदि $\vec{A}(t)$ और $\vec{B}(t)$ सभी t के लिये अशून्य सदिश और $\vec{A}(0) = 2\hat{i}+3\hat{j}$, $\vec{A}(1) = 6\hat{i}+2\hat{j}$, $\vec{B}(0) = 3\hat{i}+2\hat{j}$ एवं $\vec{B}(1) = 2\hat{i}+6\hat{j}$ हो, तो प्रदर्शित कीजिए कि कुछ t के लिये $\vec{A}(t)$ और $\vec{B}(t)$ समान्तर है।

Ans.

Sol. $\vec{A}(t)$ is parallel to $\vec{B}(t)$ for some $t \in [0,1]$ if and only if $\frac{f_1(t)}{g_1(t)} = \frac{f_2(t)}{g_2(t)}$ for some $t \in [0,1]$

or $f_1(t).g_2(t) = f_2(t).g_1(t)$ for some $t \in [0,1]$

Let $h(t) = f_1(t).g_2(t) - f_2(t).g_1(t)$

$h(0) = f_1(0).g_2(0) - f_2(0).g_1(0)$

$= 2 \times 2 - 3 \times 3 = -5 < 0$

$h(1) = f_1(1).g_2(1) - f_2(1).g_1(1)$

$= 6 \times 6 - 2 \times 2 = 32 > 0$

Since h is a continuous function, and $h(0).h(1) < 0$

$\Rightarrow$ there is some $t \in [0,1]$ for which $h(t)=0$ i.e., $\vec{A}(t)$ and $\vec{B}(t)$ are parallel vectors for this t.

26. Let V be the volume of the parallelopiped formed by the vectors $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$, $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$. If a_r, b_r, c_r, where r = 1, 2, 3, are non-negative real numbers and $\sum_{r=1}^{3}(a_r + b_r + c_r) = 3L$, show that $V \le L^3$.

माना सदिशों $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$, $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$ से निर्मित समान्तर षट्फलक का आयतन V है। यदि a_r, b_r, c_r, जहाँ r = 1, 2, 3, अऋणात्मक वास्तविक संख्याएँ है और $\sum_{r=1}^{3}(a_r + b_r + c_r) = 3L$, तो प्रदर्शित कीजिये कि $V \le L^3$

Ans.

Sol. Given that, $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$

$\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$

$\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$

where a_r, b_r, c_r, r = 1,2,3 are all non negative real numbers.

Also $\sum_{r=1}^{3}(a_r + b_r + c_r) = 3L$

To prove $V \le L^3$ Where V is vol. of parallelopiped formed by the vectors $\vec{a}, \vec{b}$ and $\vec{c}$

$\therefore$ We have $V = [\vec{a}\,\vec{b}\,\vec{c}] = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$

$\Rightarrow V = (a_1b_2c_3 + a_2b_3c_1 + a_3b_1c_2) - (a_1b_3c_2 + a_2b_1c_3 + a_3b_2c_1)$....(1)

Now we know that AM $\geq$ GM

$$\therefore \frac{(a_1+b_1+c_1)+(a_2+b_2+c_2)+(a_3+b_3+c_3)}{3}$$

$$\geq [(a_1+b_1+c_1)(a_2+b_2+c_2)(a_3+b_3+c_3)]^{1/3}$$

$$\Rightarrow \frac{3L}{3} \geq [(a_1+b_1+c_1)(a_2+b_2+c_2)(a_3+b_3+c_3)]^{1/3}$$

$\Rightarrow L^3 \geq (a_1+b_1+c_1)(a_2+b_2+c_2)(a_3+b_3+c_3)$

$= a_1b_2c_3 + a_2b_3c_1 + a_3b_1c_2 + 24$ more such terms

$\geq a_1b_2c_3 + a_2b_3c_1 + a_3b_1c_2$

$[\because a_r, b_r, c_r \geq 0$ or $r = 1,2,3]$

$\geq (a_1b_2c_3 + a_2b_3c_1 + a_3b_1c_2)$

$- (a_1b_3c_2 + a_2b_1c_3 + a_3b_2c_1)$ [same reason]

= V from (1)

Thus, $L^3 \geq V$ Hence Proved

27. If $\vec{u}, \vec{v}, \vec{w}$ are three non-coplanar unit vectors and α, β, γ are the angles between $\vec{u}$ and $\vec{v}$, $\vec{v}$ and $\vec{w}$, $\vec{w}$ and $\vec{u}$ respectively and $\vec{x}, \vec{y}, \vec{z}$ are unit vectors along the bisectors of the angles α, β, γ respectively. Prove that

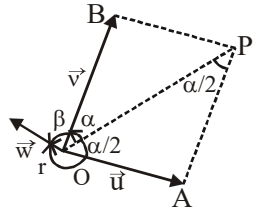
$$\left[\vec{x}\times\vec{y}\ \ \vec{y}\times\vec{z}\ \ \vec{z}\times\vec{x}\right] = \frac{1}{16}\left[\vec{u}\ \vec{v}\ \vec{w}\right]^2 \sec^2\frac{\alpha}{2}\ \sec^2\frac{\beta}{2}\ \sec^2\frac{\gamma}{2}$$

यदि $\vec{u}, \vec{v}, \vec{w}$ तीन विषमतलीय इकाई सदिश है और α, β, γ क्रमशः $\vec{u}$ और $\vec{v}$, $\vec{v}$ और $\vec{w}$, $\vec{w}$ और $\vec{u}$ के बीच के कोण है और $\vec{x}, \vec{y}, \vec{z}$ क्रमशः α, β, γ के लम्बअर्द्धको की दिशा में इकाई सदिश है। सिद्ध कीजिये कि

$$\left[\vec{x}\times\vec{y}\ \ \vec{y}\times\vec{z}\ \ \vec{z}\times\vec{x}\right] = \frac{1}{16}\left[\vec{u}\ \vec{v}\ \vec{w}\right]^2 \sec^2\frac{\alpha}{2}\ \sec^2\frac{\beta}{2}\ \sec^2\frac{\gamma}{2}$$

Ans.

Sol. Given that u, ν, ω are three non coplanar unit vectors. Angle between $\vec{u}$ and $\vec{v}$ is α, between $\vec{v}$ and $\vec{\omega}$ is β and between $\vec{\omega}$ and $\vec{u}$ it is γ. In fig. $\overrightarrow{OA}$ and $\overrightarrow{OB}$ represent $\vec{u}$ and $\vec{v}$. Let P be a pt. on angle bisector of $\angle AOB$ such that OAPB is a parallelogram.



Also $\angle POA = \angle BOP = \alpha/2$

$\therefore \angle APO = \angle BOP = \alpha/2$ (Alternate angles)

$\therefore$ In Δ OAP, OA = AP

$\therefore \quad \overrightarrow{OP} = \overrightarrow{OA} + \overrightarrow{AP} = \vec{u} + \vec{v}$

$\therefore \quad \overrightarrow{OP} = \dfrac{\vec{u}+\vec{v}}{|\vec{u}+\vec{v}|}$ i.e. $\vec{x} = \dfrac{\vec{u}+\vec{v}}{|\vec{u}+\vec{v}|}$

But $|\vec{u}+\vec{v}|^2 = (\vec{u}+\vec{v}).(\vec{u}+\vec{v})$

$= 1 + 1 + 2\,\vec{u}.\vec{v}$

$[\because \ |\vec{u}| = |\vec{v}| = 1]$

$= 2 + 2\cos\alpha = 4\cos^2\alpha/2.$

$\therefore \ |\vec{u}| = |\vec{v}| = 2\cos\alpha/2$

$\Rightarrow \quad \vec{x} = \dfrac{1}{2}\sec(\alpha/2)\ (\vec{u}+\vec{v})$

Similarly, $\vec{y} = \dfrac{1}{2}\sec(\beta/2)\ (\vec{v}+\vec{\omega})$

$$\vec{z} = \frac{1}{2}\sec(\gamma/2)(\vec{\omega}+\vec{u})$$

Now consider $[\vec{x}\times\vec{y}\ \ \vec{y}\times\vec{z}\ \ \vec{z}\times\vec{x}]$

$= (\vec{x}\times\vec{y}).[(\vec{y}\times\vec{z})\times(\vec{z}\times\vec{x})]$

$= (\vec{x}\times\vec{y}).[\{(\vec{y}\times\vec{z}).\vec{x}\}\vec{z} - \{(\vec{y}\times\vec{z}).\vec{z}\}\vec{x}]$

[Using defⁿ of vector triple product,]

$= (\vec{x}\times\vec{y}).[[\vec{x}\,\vec{y}\,\vec{z}]\vec{z} - 0]$

$= [\vec{x}\,\vec{y}\,\vec{z}][\vec{x}\,\vec{y}\,\vec{z}] \quad [\because \ [\vec{y}\,\vec{z}\,\vec{z}] = 0]$

$= [\vec{x}\,\vec{y}\,\vec{z}]^2 \qquad ...(i)$

Also $\quad [\vec{x}\,\vec{y}\,\vec{z}] = \left[\dfrac{1}{2}\sec\dfrac{\alpha}{2}(\vec{u}+\vec{v}) \ \ \dfrac{1}{2}\sec\dfrac{\beta}{2}\right.$

$$\left.(\vec{v}+\vec{\omega}) \ \ \frac{1}{2}\sec(\gamma/2)(\vec{w}+\vec{u}))\right]$$

$$= \frac{1}{8}\sec(\alpha/2)\sec(\beta/2)\sec(\gamma/2)[\vec{u}+\vec{v}\ \ \vec{v}+\vec{\omega}\ \ \vec{\omega}+\vec{u}]$$

$$= \frac{1}{8}\sec(\alpha/2)\sec(\beta/2)\sec(\gamma/2)$$

$$[(\vec{u}+\vec{v}).(\vec{v}+\vec{\omega})\times(\vec{\omega}+\vec{u})]$$

$$= \frac{1}{8}\sec(\alpha/2)\sec(\beta/2)\sec(\gamma/2)$$

$$[(\vec{u}+\vec{v}).(\vec{v}\times\vec{\omega}+\vec{v}\times\vec{u}+\vec{\omega}\times\vec{u})]$$

$$= \frac{1}{8}\sec(\alpha/2)\sec(\beta/2)\sec(\gamma/2)[\vec{u}.\vec{v}\times\vec{\omega}+\vec{v}.\vec{\omega}\times\vec{u}]$$

$(\because\ [\vec{a}\,\vec{b}\,\vec{c}]=0$ when ever any two vectors are same)

$$= \frac{1}{8}\sec(\alpha/2)\sec(\beta/2)\sec(\gamma/2)\,2[\vec{u}\,\vec{v}\,\vec{\omega}]$$

$$= \frac{1}{4}\sec(\alpha/2)\sec(\beta/2)\sec(\gamma/2)\,2[\vec{u}\,\vec{v}\,\vec{\omega}]$$

$$\therefore\ [\vec{x}\,\vec{y}\,\vec{z}]^2 = \frac{1}{16}[\vec{u}\,\vec{v}\,\vec{\omega}]^2\sec^2\alpha/2\ \sec^2\beta/2\ \sec^2\gamma/2$$

....(ii)

From (i) and (ii),

$$[\vec{x}\times\vec{y}\quad\vec{y}\times\vec{z}\quad\vec{z}\times\vec{x}]$$

$$= \frac{1}{16}[\vec{u}\,\vec{v}\,\vec{\omega}]^2\qquad\sec^2\alpha/2\ \sec^2\beta/2\ \sec^2\gamma/2.$$

28. $\vec{a},\vec{b},\vec{c}$ and $\vec{d}$ are the position vectors of the points A ≡ (x, y, z) ; B ≡ (y, – 2z, 3x) ; C ≡ (2z, 3x, – y) and D ≡ (1, –1, 2) respectively. If $|\vec{a}| = 2\sqrt{3}$; $(\vec{a}\,\hat{\wedge}\,\vec{b}) = (\vec{a}\,\hat{\wedge}\,\vec{c})$; $(\vec{a}\,\hat{\wedge}\,\vec{d}) = \frac{\pi}{2}$ and $(\vec{a}\,\hat{\wedge}\,\hat{j})$ is obtuse, then find x, y, z.

यदि A ≡ (x, y, z) ; B ≡ (y, – 2z, 3x) ; C ≡ (2z, 3x, – y) एवं D≡(1,–1, 2) की स्थिति सदिश क्रमशः $\vec{a},\vec{b},\vec{c}$ एवं $\vec{d}$ है यदि $|\vec{a}| = 2\sqrt{3}$; $(\vec{a}\,\hat{\wedge}\,\vec{b}) = (\vec{a}\,\hat{\wedge}\,\vec{c})$; $(\vec{a}\,\hat{\wedge}\,\vec{d}) = \frac{\pi}{2}$ तथा $(\vec{a}\,\hat{\wedge}\,\hat{j})$ अधिक कोणीय है तो x, y, z का मान ज्ञात कीजिए।

Ans. x = 2, y = – 2, z = – 2

Sol. A (x, y, z) $\qquad |\vec{a}| = 2\sqrt{3}$

B (y, –2z, 3x)

C (2z, 3x, –y)

D (1, –1, 2)

$(\vec{a}\,\hat{\wedge}\,\vec{b}) = (\vec{a}\,\hat{\wedge}\,\vec{c}) = \theta$

$(\vec{a}\,\hat{\wedge}\,\vec{d}) = \frac{\pi}{2}$

$(\hat{a}\,\hat{\wedge}\,\hat{j}) > 90^o$

$\vec{a}\cdot\vec{d} = 0$

$x - y + 2z = 0$ ...(i)

$|\vec{a}|^2 = x^2 + y^2 + z^2$

$|\vec{b}|^2 = y^2 + 4z^2 + 9x^2$

$|\vec{c}| = 4z^2 + 9x^2 + y^2$

$|\vec{b}| = |\vec{c}|$

$$\cos\theta = \frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|} = \frac{\vec{a}\cdot\vec{c}}{|\vec{a}||\vec{c}|}$$

$xy - 2yz + 3xz = 2xz + 3xy - yz$
$2xy + yz - xz = 0$ ...(ii)
$x^2 + y^2 + z^2 = 12$ ...(iii)
$x - y + 2z = 0$
$y = x + 2z$
$2xy + yz - xz = 0$
$(2x + z)\, y = xz$
$(2x + z)\,(x + 2z) = xz$
$2x^2 + 5xz + 2z^2 = xz$
$2x^2 + 4xz + 2z^2 = 0$
$x^2 + 2xz + z^2 = 0$
$x = -z$
$y = z$
$x = -y = -z$
$x^2 + y^2 + z^2 = 12$
$3x^2 = 12$
$x = \pm 2$
$\vec{a} \wedge \vec{j}$ = obtuse $z < 0$
$x = 2, z = -2, y = -2$

29. If $\vec{r}$ and $\vec{s}$ are non zero constant vectors and the scalar b is chosen such that $|\vec{r} + b\vec{s}|$ is minimum, then show that the value of $|b\vec{s}|^2 + |\vec{r} + b\vec{s}|^2$ is equal to $|\vec{r}|^2$.

यदि $\vec{r}$ एवं $\vec{s}$ अशून्य तथा स्थिर सदिश है। एक अदिश b ऐसा चुना जाता है कि $|\vec{r} + b\vec{s}|$ न्यूनतम है तो सिद्ध करो कि $|b\vec{s}|^2 + |\vec{r} + b\vec{s}|^2$ का मान $|\vec{r}|^2$ होगा।

Ans.

Sol. $|\vec{r} + b\vec{s}|$ is minimum

Let $f(b) = \sqrt{\vec{r}^2 + \vec{b}^2\vec{s}^2 + 2\vec{r}.b\vec{s}}$

for maxima & minima

$$f'(b) = \frac{2b\vec{s}^2 + 2\vec{r}.\vec{s}}{\sqrt{\vec{r}^2 + b^2\vec{s}^2 + 2b\vec{r}.\vec{s}}} = 0$$

$$b = -\frac{\vec{r}.\vec{s}}{\vec{s}^2}$$

$$\left|b\vec{s}\right|^2 + \left|\vec{r} + b\vec{s}\right|^2 = b^2\vec{s}^2 + \vec{r}^2 + b^2\vec{s}^2 + 2b\vec{r}.\vec{s}$$

$$= 2b^2\vec{s}^2 + \vec{r}^2 - 2b^2\vec{s}^2 = |\vec{r}|^2$$

30. Given four non zero vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ and $\vec{d}$. The vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ are coplanar but not collinear pair by pair and vector $\vec{d}$ is not coplanar with vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ and $(\hat{\vec{a}\vec{b}}) = (\hat{\vec{b}\vec{c}}) = \frac{\pi}{3}, (\hat{\vec{d}\vec{a}}) = \alpha, (\hat{\vec{d}\vec{b}}) = \beta$ then prove that $(\hat{\vec{d}\,\vec{c}}) = \cos^{-1}(\cos\beta - \cos\alpha)$.

चार अशून्य सदिश $\vec{a}$, $\vec{b}$, $\vec{c}$ तथा $\vec{d}$ दिया हुआ है यदि सदिश $\vec{a}$, $\vec{b}$ तथा $\vec{c}$ समतलीय है लेकिन युग्म में संरेखीय नही है तथा $\vec{d}$ सदिश $\vec{a}$, $\vec{b}$ एवं $\vec{c}$ से समतलीय नही है। यदि $(\hat{\vec{a}\vec{b}}) = (\hat{\vec{b}\vec{c}}) = \frac{\pi}{3}, (\hat{\vec{d}\vec{a}}) = \alpha, (\hat{\vec{d}\vec{b}}) = \beta$ तो सिद्ध करो कि $(\hat{\vec{d}\,\vec{c}}) = \cos^{-1}(\cos\beta - \cos\alpha)$

Ans.

Sol. $\vec{a} \wedge \vec{b} = \vec{b} \wedge \vec{c} = \frac{\pi}{3}$

$\vec{d} \wedge \vec{a} = \alpha$

$\vec{d} \wedge \vec{b} = \beta$

$\vec{a}, \vec{b}, \vec{c}$ coplanar but not collinear.

$\vec{d}$ not coplanar with $\vec{a}, \vec{b}, \vec{c}$.

Let $\vec{d} \wedge \vec{c} = \theta$

$\vec{c} = x\vec{a} + y\vec{b}$

dot with $\vec{b}$

$\vec{b}\cdot\vec{c} = x\vec{a}\cdot\vec{b} + y\left|\vec{b}\right|^2$

dot with $\vec{d}$

$\vec{d}\cdot\vec{c} = x\vec{d}\cdot\vec{a} + y\vec{d}\cdot\vec{b}$

$\left|\vec{d}\right|\left|\vec{c}\right|\cos\gamma = x\left|\vec{d}\right|\left|\vec{a}\right|\cos\alpha + y\left|\vec{d}\right|\left|\vec{b}\right|\cos\beta$

$$\cos\gamma = x\frac{\left|\vec{a}\right|}{\left|\vec{c}\right|}\cos\alpha + y\frac{\left|\vec{b}\right|}{\left|\vec{c}\right|}\cos\beta \qquad ...(1)$$

dot with $\vec{b}$

$\vec{b}\cdot\vec{c} = x\vec{b}\cdot\vec{a} + y\vec{b}\cdot\vec{b}$

$$\left|\vec{b}\right|\left|\vec{c}\right|\cos\frac{\pi}{3} = x\left|\vec{b}\right|\left|\vec{a}\right|\cos\frac{\pi}{3} + y\left|\vec{b}\right|\left|\vec{b}\right|$$

$$\frac{1}{2} = \frac{x}{2}\frac{\left|\vec{a}\right|}{\left|\vec{c}\right|} + \frac{y\left|\vec{b}\right|}{\left|\vec{c}\right|} \qquad ...(2)$$

dot with $\vec{c}$

$$\left|\vec{c}\right|\left|\vec{c}\right| = x\left|\vec{a}\right|\left|\vec{c}\right|\cos\frac{2\pi}{3} + y\left|\vec{b}\right|\left|\vec{c}\right|\cos\frac{\pi}{3}$$

$$2 = -\frac{x\left|\vec{a}\right|}{\left|\vec{c}\right|} + \frac{y\left|\vec{b}\right|}{\left|\vec{c}\right|} \qquad ...(3)$$

from equation (2) & (3)

$$3 = \frac{3y\left|\vec{b}\right|}{\left|\vec{c}\right|}$$

$$\frac{y\left|\vec{b}\right|}{\left|\vec{c}\right|} = 1$$

$$\frac{x\left|\vec{a}\right|}{\left|\vec{c}\right|} = -1$$

Put these in equation (1)

$\cos\gamma = -\cos\alpha + \cos\beta$

$\cos\gamma = \cos\beta - \cos\alpha$

$\gamma = \cos^{-1}\left(\cos\beta - \cos\alpha\right)$

31. The position vectors of the vertices A, B and C of a tetrahedron are (1, 1, 1), (1, 0, 0) and (3, 0, 0) respectively. The altitude from the vertex D to the opposite face ABC meets the median line through A of the triangle ABC at a point E. If the length of side AD is 4 and volume of the tetrahedron is $\frac{2\sqrt{2}}{3}$ then find the all possible position vectors of the point E.

किसी चतुष्फलक के शीर्ष A, B एवं C की स्थिति सदिश क्रमशः (1, 1, 1), (1, 0, 0) तथा (3, 0, 0) है यदि शीर्ष D से विपरीत तल ABC पर लम्ब डाला जाता है तथा यह त्रिभुज ABC के शीर्ष A से जाने वाली माध्यिका पर बिन्दु E पर मिलता है यदि AD की लम्बाई 4 तथा चतुष्फलक का आयतन $\frac{2\sqrt{2}}{3}$ है तो E की स्थिति सदिश बताओं।

Ans. $(-1, 3, 3)$ & $(3, -1, -1)$

Sol. As, F is the mid point of BC. So, $\vec{F} = 2\hat{i}$

As, AE is perpendicular to DE. Let E divides AF in the ratio $\lambda : 1$

$$\vec{E} = \frac{2\lambda\hat{i} + 1(\hat{i} + \hat{j} + \hat{k})}{\lambda + 1} = \frac{2\lambda + 1}{\lambda + 1}\hat{i} + \frac{1}{\lambda + 1}\hat{j} + \frac{1}{\lambda + 1}\hat{k}$$

$$\text{Volume of tetrahedron} = 1/3 \times \text{Area of base} \times \text{Height} = \frac{2\sqrt{2}}{3}$$

$$\text{Area of base} = \frac{1}{2}|\overrightarrow{BC} X \overrightarrow{BA}| = \sqrt{2}$$

$$\frac{1}{3} \cdot \sqrt{2} \cdot DE = 2\frac{\sqrt{2}}{3}$$

Hence $DE = 2$

As $\angle ADE$ is right angle

So $AD^2 = AE^2 + DE^2$

$AE^2 = 12$

$$\overrightarrow{AE} = \vec{E} - \vec{A} = \frac{\lambda}{\lambda + 1}\hat{i} - \frac{\lambda}{\lambda + 1}\hat{j} - \frac{\lambda}{\lambda + 1}\hat{k}$$

$$|\overrightarrow{AE}|^2 = 12$$

$$3\frac{\lambda^2}{(\lambda + 1)^2} = 12$$

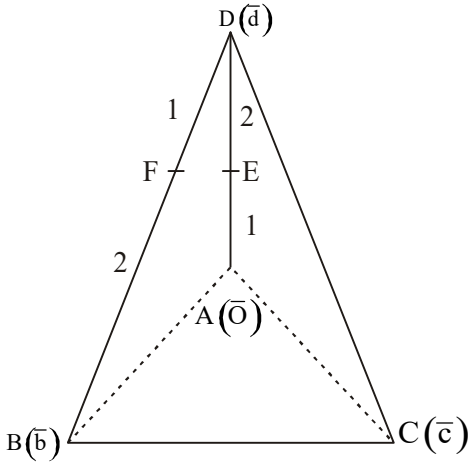
On solving $\lambda = -2, -\frac{2}{3}$

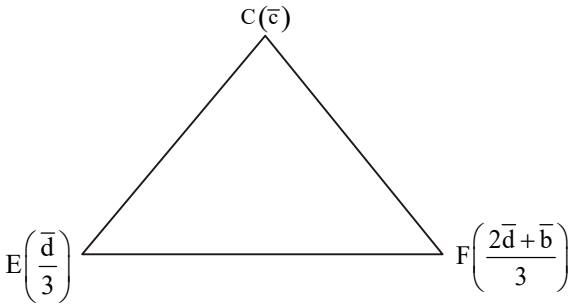
32. The length of the edge of the regular tetrahedron $D-ABC$ is 'a'. Point E and F are taken on the edges AD and BD respectively such that E divides $\vec{DA}$ and F divides $\vec{BD}$ in the ratio 2:1 each. Then find the area of triangle CEF.

एक समचतुष्फलक D – ABC की भुजा 'a' है यदि भुजा AD एवं BD पर बिन्दु E तथा F इस प्रकार से लिया जा रहा है कि fcUnq E $\vec{DA}$ को तथा बिन्दु F सदिश $\vec{BD}$ को 2:1 के अनुपात में बाट रही है तो त्रिभुज CEF का क्षेत्रफल बताइये।

Ans. $\frac{5a^2}{12\sqrt{3}}$ sq. units

Sol.





$c(\bar{c}), E\left(\frac{\bar{d}}{3}\right)$ and $F\left(\frac{2\bar{d}+\bar{b}}{3}\right)$

$\bar{c} \cap \bar{d} = \bar{b} \wedge \bar{d} = \bar{b} \wedge \bar{c} = 60$

$$\overline{CE} = \left|\frac{\bar{d}}{3} - \bar{c}\right| = \sqrt{\frac{a^2}{9} + a^2 - \frac{2}{3}a^2 \cdot \frac{1}{2}} = \frac{\sqrt{7}a}{3}$$

$$\overline{CF} = \left|\frac{2\bar{d}+\bar{b}}{3} - \bar{c}\right| = \sqrt{\frac{4a^2}{9} + \frac{a^2}{9} + a^2 + 2\left(\frac{2}{9}a^2 \cdot \frac{1}{2} - \frac{a^2}{3} \cdot \frac{1}{2} - \frac{2a^2}{3} \cdot \frac{1}{2}\right)}$$

$$= \frac{\sqrt{7}a}{3}$$

$$\overline{EF} = \left|\frac{\bar{b}+\bar{d}}{3}\right| = \frac{1}{3}\sqrt{a^2 + a^2 + 2 \cdot a^2 \cdot \frac{1}{2}} = \frac{\sqrt{3}a}{3} = \frac{a}{\sqrt{3}}$$

Now, use heron's formula for area.

33. Given three points on the xy plane on O(0, 0), A(1, 0) and B(–1, 0). Point P is moving on the plane satisfying the *condition* $\left(\overrightarrow{PA}\cdot\overrightarrow{PB}\right)+3\left(\overrightarrow{OA}\cdot\overrightarrow{OB}\right)=0$. If the maximum and minimum values of $\left|\overrightarrow{PA}\right|\left|\overrightarrow{PB}\right|$ are M and *m* respectively then find the value of $M^2 + m^2$.

गल तल पर O(0, 0), A(1, 0) तथा B(–1, 0) दिया हुआ है इसी तल पर एक बिन्दु P इस प्रकार से गति कर रहा है कि $\left(\overrightarrow{PA}\cdot\overrightarrow{PB}\right)+3\left(\overrightarrow{OA}\cdot\overrightarrow{OB}\right)=0$. यदि $\left|\overrightarrow{PA}\right|\left|\overrightarrow{PB}\right|$ का अधिकतम तथा न्यूनतम मान M एवं *m* है तो $M^2 + m^2$ का मान बताओं।

Ans. 34

Sol. O (0,0), A(1,0) & B (–1, 0)

Let P (x,y)

$\overrightarrow{PA} = (1-x)\hat{i} - y\hat{j}$

$\overrightarrow{PB} = -(1+x)\hat{i} - y\hat{j}$

$\overrightarrow{PA}.\overrightarrow{PB} + 3\ \overrightarrow{OA}.\overrightarrow{OB} = 0$

$\Rightarrow (x^2 - 1) + y^2 - 3 = 0$

$x^2 + y^2 = 4$ _____(1)

$\left|\overrightarrow{PA}\right|\left|\overrightarrow{PB}\right| = \sqrt{(x-1)^2+y^2}\sqrt{(x+1)^2+y^2}$

$= \sqrt{5-2x}\cdot\sqrt{5+2x}$

$= \sqrt{25-4x^2}$, $x\in(-2,2)$ (from (1))

so M = 5, m = 3

$\Rightarrow M^2 + m^2 = 25 + 9 = 34$

Paragraph for Question Nos. 34 to 36

Let $|\vec{a}| = |\vec{b}| = 2$ and $|\vec{c}| = 1$. Also $(\vec{a}-\vec{c})\cdot(\vec{b}-\vec{c}) = 0$ and $|\vec{a}-\vec{b}|^2 + |\vec{a}+\vec{b}|^2 = 16$

माना $|\vec{a}| = |\vec{b}| = 2$ तथा $|\vec{c}| = 1$ साथ ही $(\vec{a}-\vec{c})\cdot(\vec{b}-\vec{c}) = 0$ तथा $|\vec{a}-\vec{b}|^2 + |\vec{a}+\vec{b}|^2 = 16$

34. $\left|\vec{a}-\vec{b}\right|^2 + 2\vec{c}\cdot\left(\vec{a}+\vec{b}\right)$ has the value equal to :

$\left|\vec{a}-\vec{b}\right|^2 + 2\vec{c}\cdot\left(\vec{a}+\vec{b}\right)$ का मान बराबर होगा :

(A) 12 (B) 10 (C) 8 (D) 6

Ans. B

35. $\left|\vec{a}+\vec{b}-\vec{c}\right|^2$ equals :

$\left|\vec{a}+\vec{b}-\vec{c}\right|^2$ बराबर है :

(A) 5 (B) 6 (C) 7 (D) 8

Ans. C

36. Difference of the maximum and minimum value of $\left|\vec{a}+\vec{b}\right|$ is equal to :

$\left|\vec{a}+\vec{b}\right|$ के अधिकतम एवं न्यूनतम मान का अंतर है :

(A) 2 (B) 3 (C) 4 (D) 1

Ans. A

Sol. Given $\left|\vec{a}-\vec{b}\right|^2+\left|\vec{a}+\vec{b}\right|^2=16$ ...(1) ; $|\vec{a}|=|\vec{b}|=2$ and $|\vec{c}|=1$

Also, $\left|\vec{a}-\vec{b}\right|^2=\left|(\vec{a}-\vec{c})-(\vec{b}-\vec{c})\right|^2$

$\left|\vec{a}-\vec{b}\right|^2=(\vec{a}-\vec{c})^2+(\vec{b}-\vec{c})^2-\underbrace{2(\vec{a}-\vec{c})\cdot(\vec{b}-\vec{c})}_{\text{zero}}$

$=\vec{a}^2+\vec{c}^2-2\vec{a}\cdot\vec{c}+\vec{b}^2+\vec{c}^2-2\vec{b}\cdot\vec{c}$

$\left|\vec{a}-\vec{b}\right|^2=10-2\left(\vec{a}\cdot\vec{c}+\vec{b}\cdot\vec{c}\right)$ ...(2)

$\Rightarrow \left|\vec{a}-\vec{b}\right|^2+2\vec{c}\cdot\left(\vec{a}+\vec{b}\right)=10 \Rightarrow$ B Ans.

subsituting $\left|\vec{a}-\vec{b}\right|^2=16-\left|\vec{a}+\vec{b}\right|^2$ from (1) in (2)

we get, $16-\left|\vec{a}+\vec{b}\right|^2=10-2\left(\vec{a}\cdot\vec{c}+\vec{b}\cdot\vec{c}\right)$

$\left|\vec{a}+\vec{b}\right|^2-2\vec{c}\cdot(\vec{a}+\vec{b})=6$ .....(3)

$\left|\vec{a}+\vec{b}\right|^2-2\vec{c}\cdot(\vec{a}+\vec{b})+\vec{c}^2=6+\vec{c}^2=7$

$\left|\vec{a}+\vec{b}-\vec{c}\right|^2=7 \Rightarrow$ C

Again, using triangle inequality, we have

$\left|\vec{a}+\vec{b}\right|-\left|\vec{c}\right| \le \left|\vec{a}+\vec{b}-\vec{c}\right| \le \left|\vec{a}+\vec{b}\right|+\left|\vec{c}\right|$

Hence $\left|\vec{a}+\vec{b}-\vec{c}\right|-\left|\vec{c}\right| \le \left|\vec{a}+\vec{b}\right| \le \left|\vec{a}+\vec{b}-\vec{c}\right|+\left|\vec{c}\right|$

$\sqrt{7}-1 \le \left|\vec{a}+\vec{b}\right| \le \sqrt{7}+1$

$\therefore \left|\vec{a}+\vec{b}\right|_{max} = \sqrt{7}+1$

and $\left|\vec{a}+\vec{b}\right|_{min} = \sqrt{7}-1$

$\therefore$ Difference $= 2 \Rightarrow$ A

Note: $\sqrt{16-(8+2\sqrt{7})} \le \left|\vec{a}-\vec{b}\right| \le \sqrt{16-(8-2\sqrt{7})}$

$\Rightarrow \sqrt{7}-1 \le \left|\vec{a}-\vec{b}\right| \le 1+\sqrt{7}$

Paragraph for Question Nos. 37 to 39

Consider the three vectors $\vec{p}$, $\vec{q}$ and $\vec{r}$ such that

$\vec{p}=\hat{i}+\hat{j}+\hat{k}$; $\vec{q}=\hat{i}-\hat{j}+\hat{k}$

$\vec{p}\times\vec{r}=\vec{q}+c\vec{p}$ and $\vec{p}\cdot\vec{r}=2$

माना कि सदिश $\vec{p}$, $\vec{q}$ तथा $\vec{r}$

$\vec{p}=\hat{i}+\hat{j}+\hat{k}$; $\vec{q}=\hat{i}-\hat{j}+\hat{k}$

$\vec{p}\times\vec{r}=\vec{q}+c\vec{p}$ तथा $\vec{p}\cdot\vec{r}=2$

37. The value of $\left[\vec{p}\ \vec{q}\ \vec{r}\right]$ is :

(A) $-\dfrac{5\sqrt{2}c}{|\vec{r}|}$ (B) $-\dfrac{8}{3}$ (C) 0 (D) greater then zero

$\left[\vec{p}\ \vec{q}\ \vec{r}\right]$ का मान होगा :

(A) $-\dfrac{5\sqrt{2}c}{|\vec{r}|}$ (B) $-\dfrac{8}{3}$ (C) 0 (D) 0 से अधिक

Ans. B

Sol. Given $\vec{p}=\hat{i}+\hat{j}+\hat{k}$

$\vec{q}=\hat{i}-\hat{j}+\hat{k}$

$\vec{p}\times\vec{r}=\vec{q}+c\vec{p}$ and $\vec{p}\,.\,\vec{r}=2$

cross with $\vec{p}$ dot with $\vec{p}$

$\therefore \quad \vec{p} \times (\vec{p} \times \vec{r}) = \vec{p} \times (\vec{q} + c\vec{p})$ and $\vec{p} \cdot (\vec{p} \times \vec{r}) = \vec{p} \cdot (\vec{q} + c\vec{p})$

$\therefore \quad (\vec{p} \cdot \vec{r})\vec{p} - (\vec{p} \cdot \vec{p})\vec{r} = \vec{p} \times \vec{q} + c\vec{0}$ $0 = \vec{p} \cdot \vec{q} + c\,(\vec{p} \cdot \vec{p})$

$\therefore \quad (\vec{p} \cdot \vec{p})\vec{r} = (\vec{p} \cdot \vec{r})\,\vec{p} - \vec{p} \times \vec{q}$ ... (i) $c = -\dfrac{\vec{p} \cdot \vec{q}}{\vec{p} \cdot \vec{p}}$ ... (ii)

But $\vec{p} \cdot \vec{p} = |\vec{p}| = p^2 = 3$ ... (iii)

$\vec{p} \cdot \vec{q} = 1 - 1 + 1 = 1$ ... (iv)

$$\vec{p} \times \vec{q} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 1 & -1 & 1 \end{vmatrix} = 2\hat{i} - 2\hat{k} \quad \text{... (v)}$$

using (iii), (iv), (v) in (i) and (ii), we get

$3\vec{r} = 2\vec{p} - 2i + 2k$ and $c = -\dfrac{1}{3}$ ....(vii)

$\therefore \quad \vec{r} = \dfrac{1}{3}\left[2\hat{i} + 2\hat{j} + 2\hat{k} - 2\hat{i} + 2\hat{k}\right]$

$\therefore \quad \vec{r} = \dfrac{2}{3}(\hat{j} + 2\hat{k})$ ....(vi)

Now $$[\vec{p}\ \vec{q}\ \vec{r}] = \begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 0 & \frac{1}{2} & \frac{4}{3} \end{vmatrix} = \left(\frac{-4}{3} - \frac{2}{3}\right) - 1\left(\frac{4}{3}\right) + 1\left(\frac{2}{3}\right)$$

$\therefore \quad [\vec{p}\ \vec{q}\ \vec{r}] = -2 - \dfrac{2}{5} = -\dfrac{8}{3}$ ... (viii)

38. If $\vec{x}$ is a vector such that $[\vec{p}\,\vec{q}\,\vec{r}]\vec{x} = (\vec{p} \times \vec{q}) \times \vec{r}$, then $\vec{x}$ is :

(A) $c\left(\hat{i} - 2\hat{j} + \hat{k}\right)$ (B) a unit vector

(C) indeterminate, as $[\vec{p}\,\vec{q}\,\vec{r}]$ (D) $-\dfrac{1}{2}(\hat{i} - 2\hat{j} + \hat{k})$

यदि $\vec{x}$ इस प्रकार है कि $[\vec{p}\,\vec{q}\,\vec{r}]\vec{x} = (\vec{p} \times \vec{q}) \times \vec{r}$, तो $\vec{x}$ है :

(A) $c\left(\hat{i} - 2\hat{j} + \hat{k}\right)$ (B) एक इकाई सदिश

(C) $[\vec{p}\,\vec{q}\,\vec{r}]$ के समान अनिरधार्य (D) $-\dfrac{1}{2}(\hat{i} - 2\hat{j} + \hat{k})$

Ans. D

Sol. $[\vec{p}\ \vec{q}\ \vec{r}]\vec{x} = (\vec{p}\times\vec{q})\times\vec{r}$

$\Rightarrow \quad \left(-\frac{8}{3}\right)\vec{x} = (\vec{p}\cdot\vec{r})\vec{q} - (\vec{q}\cdot\vec{r})\vec{p}$

or $\quad \left(-\frac{8}{3}\right)\vec{x} = 2\vec{q} - \frac{2\vec{p}}{3} \quad \left(\vec{q}\cdot\vec{r} = \frac{2}{3} \text{ Verify your self}\right)$

$\therefore \quad \vec{x} = -\frac{3}{8}\cdot\frac{2}{3}(3\vec{q} - \vec{p}) = -\frac{1}{4}(3\hat{i} - 3\hat{j} + 3\hat{k} - \hat{i} - \hat{j} - \hat{k}) = -\frac{1}{4}(2\vec{i} - 4\vec{j} + 2\vec{k})$

$\therefore \quad \vec{x} = -\frac{1}{2}(\hat{i} - 2\hat{j} + \hat{k})$

39. If $\vec{y}$ is a vector satisfying $(1 + c)\vec{y} = \vec{p}\times(\vec{q}\times\vec{r})$ then the vectors $\vec{x}, \vec{y}, \vec{r}$:

(A) are collinear

(B) are coplanar

(C) represent the coterminus edges of a tetrahedron whose volume is | c | cubic units.

(D) represent the coterminus edges of a parallelepiped whose volume is | c | cubic units

यदि $\vec{y}$ एक सदिश है जो $(1 + c)\vec{y} = \vec{p}\times(\vec{q}\times\vec{r})$ को संतुष्ट करता है, तो सदिश $\vec{x}, \vec{y}, \vec{r}$ है :

(A) समरेखीय

(B) समतलीय

(C) एक चतुष्फलक की संगामी भुजाऐं दर्शाते है जिसका आयतन | c | घन इकाई है

(D) एक समान्तर षट्फलक की संगामी भुजाऐं दर्शाते है जिसका आयतन | c | घन इकाई है

Ans. C

Sol. As $\quad c = -\frac{1}{3}$ from (vii)

$\therefore \quad \left(1 - \frac{1}{3}\right)\vec{y} = (\vec{p}\cdot\vec{r})\vec{q} - (\vec{p}\cdot\vec{q})\vec{r}$

$\therefore \quad \frac{2}{3}\vec{y} = 2\vec{q} - (1)\vec{r} \qquad (\text{As } \vec{p}\cdot\vec{q} = 1 \text{ from .... (iv)})$

$\therefore \quad \vec{y} = \frac{3}{2}\left(2\hat{i} - 2\hat{j} + 2\hat{k} - \frac{2}{3}\hat{j} - \frac{4}{3}\hat{k}\right) = \frac{3}{2}\,\frac{(6\hat{i} - 8\hat{j} + 2\hat{k})}{3}$

$\vec{y} = 3\hat{i} - 4\hat{j} + \hat{k}$

$\therefore \quad [\vec{x}\ \vec{y}\ \vec{r}] = \begin{vmatrix} -\frac{1}{2} & 1 & -\frac{1}{2} \\ 3 & -4 & 1 \\ 0 & \frac{2}{3} & \frac{4}{3} \end{vmatrix} = -\frac{1}{2}\left(-\frac{16}{3} - \frac{2}{3}\right) - 1\,(4) - \frac{1}{2}\,(2)$

$\therefore \quad [\vec{x}\ \vec{y}\ \vec{r}\,] = 3 - 4 - 1 = -2$

$\therefore \quad \left|\frac{1}{6}[\vec{x}\ \vec{y}\ \vec{r}\,]\right| = \left|-\frac{1}{3}\right| = |c|$

$\Rightarrow$ $\vec{x}$, $\vec{y}$, $\vec{r}$ are the coterminus edges of a tetrahedron whose volume is $|c|$.

Paragraph for Question Nos. 40 to 42

Suppose the three vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ on a plane satisfy the condition that $|\vec{a}| = |\vec{b}| = |\vec{c}| = |\vec{a} + \vec{b}| = 1$;

$\vec{c}$ is perpendicular to $\vec{a}$ and $\vec{b} \cdot \vec{c} > 0$, then :

माना कि किसी तल में तीन सदिश $\vec{a}$, $\vec{b}$, $\vec{c}$ है जो कि दिये हुये प्रतिबन्धों को सन्तुष्ट करते है $|\vec{a}| = |\vec{b}| = |\vec{c}| = |\vec{a} + \vec{b}| = 1$;

यदि $\vec{c}$ सदिश $\vec{a}$ के लम्बवत् है तथा $\vec{b} \cdot \vec{c} > 0$, तो :

40. Find the angle formed by $2\vec{a} + \vec{b}$ and $\vec{b}$.

$2\vec{a} + \vec{b}$ एवं $\vec{b}$ के बीच का कोण बताइये।

Ans. $\frac{\pi}{2}$

Sol. $\left|\bar{a} + \bar{b}\right|^2 = a^2 + b^2 + 2ab\cos\theta = 1$

$\Rightarrow \cos\theta = -\frac{1}{2}$ and $\bar{a} \cdot \bar{b} = -\frac{1}{2}$

$$\cos\theta = \frac{\bar{b} \cdot \left(2\bar{a} + \bar{b}\right)}{\left|\bar{b}\right|\left|2\bar{a} + \bar{b}\right|} = \frac{-1+1}{\left|\bar{b}\right|\left|2\bar{a} + \bar{b}\right|} = 0$$

So, $\theta = \frac{\pi}{2}$

41. If the vector $\vec{c}$ is expressed as a linear combination $\lambda\vec{a} + \mu\vec{b}$ then find the ordered pair (λ, μ).

यदि सदिश $\vec{c}$ को एक घात संचय के रूप में यदि $\lambda\vec{a} + \mu\vec{b}$ से व्यक्त किया जाय तो (λ, μ) का मान बताइये।

Ans. $\left(\frac{1}{\sqrt{3}}, \frac{2}{\sqrt{3}}\right)$

Sol. $\left|\bar{a}+\bar{b}\right|^2 = a^2+b^2+2ab\cos\theta = 1$

$\Rightarrow \cos\theta = -\frac{1}{2}$ and $\bar{a}\cdot\bar{b} = -\frac{1}{2}$

$\bar{c} = \lambda\bar{a}+\mu\bar{b}$

$\bar{c}\cdot\bar{c} = \lambda a^2 + \mu\left(\bar{a}\cdot\bar{b}\right)$

$\Rightarrow 0 = \lambda - \frac{\mu}{2}$ $\Rightarrow m = 2\lambda$

and $|\bar{c}| = 1$

$\Rightarrow \lambda^2a^2 + \mu^2b^2 + 2\lambda\mu\left(\bar{a}\cdot\bar{b}\right) = 1$

$\lambda^2 + 4\lambda^2 - 2\lambda^2 = 1$

$\Rightarrow 3\lambda^2 = 1$

$\Rightarrow \lambda = \frac{1}{\sqrt{3}}$ So, $\mu = \frac{2}{\sqrt{3}}$

Note: $\bar{b}\cdot\bar{c} > 0$

$\Rightarrow \lambda\left(\bar{a}\cdot\bar{b}\right)+\mu b^2 > 0$

$\Rightarrow \mu - \frac{\lambda}{2} > 0$

$\Rightarrow \mu > \frac{\lambda}{2}$

42. For real numbers x, y the vector $\vec{p} = x\vec{a} + y\vec{c}$ satisfies the condition $0 \le \vec{p}\cdot\vec{a} \le 1$ and $0 \le \vec{p}\cdot\vec{b} \le 1$. Find the maximum value of $\vec{p}\cdot\vec{c}$.

वास्तविक संख्या x एवं y के लिये यदि सदिश $\vec{p} = x\vec{a} + y\vec{c}$ यदि शर्त $0 \le \vec{p}\cdot\vec{a} \le 1$ एवं $0 \le \vec{p}\cdot\vec{b} \le 1$ को संतुष्ट कर रहा है तो $\vec{p}\cdot\vec{c}$ अधिकतम मान बताओं।

Ans. $\sqrt{3}$

Sol. $\bar{p} = x\bar{a} + y\bar{c}$

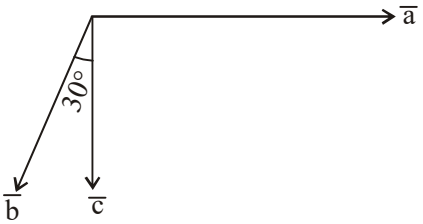
$\bar{p}\cdot\bar{a}=x \qquad \Rightarrow 0\le x\le 1$

$\bar{p}\cdot\bar{b}-x\left(\frac{-1}{2}\right)+y\left(\bar{b}\cdot\bar{c}\right) \qquad \Rightarrow 0\le -\frac{x}{2}+y\left(\bar{b}\cdot\bar{c}\right)\le 1 \qquad$ _____(i)

and $\bar{p}\cdot\bar{c}=y$

$\bar{a},\bar{b},\bar{c}$ are in same plane

$\bar{a}\cdot\bar{b}=-\frac{1}{2} \qquad \hat{a}\wedge\bar{b}=120°$



and $\bar{a}\perp\bar{c}$

From equation (i)

$$y=\frac{1+\frac{x}{2}}{\bar{b}\cdot\bar{c}}\le\frac{(2+x)}{2\left(\bar{b}\cdot\bar{c}\right)}$$

$$\Rightarrow y_{max}=\frac{2+1}{2\left(\bar{b}\cdot\bar{c}\right)} \qquad \{\because\ 0\le x\le 1\}$$

$$=\frac{3}{2\cdot\frac{\sqrt{3}}{2}}=\sqrt{3}$$

43. Match the statements/expressions given in Column I with the values given in Column II :

Column I	**Column II**
A. Root(s) of the equation $2\sin^2\theta+\sin^2 2\theta=2$	p. $\frac{\pi}{6}$
B. Points of discontinuity of the function $f(x)=\left[\frac{6x}{\pi}\right]\cos\left[\frac{3x}{\pi}\right]$, where [y] denotes the largest integer less than or equal to y	q. $\frac{\pi}{4}$
C. Volume of the parallelopiped with its edges represented by the vectors $\hat{i}+\hat{j}$, $\hat{i}+2\hat{j}$ and $\hat{i}+\hat{j}+\pi\hat{k}$	r. $\frac{\pi}{3}$
D. Angle between vectors $\vec{a}$ and $\vec{b}$ where $\vec{a},\vec{b}$ and $\vec{c}$ are unit vectors satisfying $\vec{a}+\vec{b}+\sqrt{3}\,\vec{c}=\vec{0}$	s. $\frac{\pi}{2}$
	t. π

कॉलम–I में दिए गये वक्तयों/व्यंजको का कॉलम–II में दिये गये मानों से सुमेलित कीजिए :

Column I	**Column II**
A. समीकरण $2\sin^2\theta + \sin^2 2\theta = 2$ के मूल है	p. $\frac{\pi}{6}$
B. फलन $f(x) = \left[\frac{6x}{\pi}\right]\cos\left[\frac{3x}{\pi}\right]$ के असतत्ता बिन्दु [y] उस अधिकतम पूर्णांक को दर्शाता है जो कि y के बराबर या उससे कम हो	q. $\frac{\pi}{4}$
C. सदिशो $\hat{i}+\hat{j}, \hat{i}+2\hat{j}$ तथा $\hat{i}+\hat{j}+\pi\hat{k}$ से निरूपित कोरों वाले समान्तर षट्फलक का आयतन	r. $\frac{\pi}{3}$
D. सदिशो $\vec{a}$ एवं $\vec{b}$ के बीच को कोण, जबकि $\vec{a},\vec{b}$ व $\vec{c}$, $\vec{a}+\vec{b}+\sqrt{3}\vec{c}=\vec{0}$ को संतुष्ट करने वाले मात्रक सदिश है	s. $\frac{\pi}{2}$
	t. π

Ans. (A) Q, S ; (B) P, R, S, T ; (C) T ; (D) R

Sol. (A) $\sin^2 2\theta = 2 - 2\sin^2\theta = 2\cos^2\theta$

$4\sin^2\theta\cos^2\theta = 2\cos^2\theta$

$\cos\theta = 0$ OR $\sin^2\theta = \frac{1}{2}$

(B) $\frac{6x}{\pi} = I_1 \quad \Rightarrow x = \frac{\pi I_1}{6}$

and $\frac{3x}{\pi} = I_2 \quad \Rightarrow x = \frac{I_2\pi}{3}$

$x = \frac{\pi}{6}, \frac{\pi}{3}, \frac{\pi}{2}, \pi \qquad x \neq \frac{\pi}{4}$

(C) $\text{Volume} = \left[\hat{i}+\hat{j}, \hat{i}+2\hat{j}, \hat{i}+\hat{j}+\pi\hat{k}\right] = \pi$

(D) $\bar{a}+\bar{b} = -\sqrt{3}\bar{c}$

$a^2 + b^2 + 2ab + \cos\theta = 3c^2$

$\Rightarrow \cos\theta = \frac{1}{2} \qquad \Rightarrow \theta = \frac{\pi}{3}$

44. Let $\hat{a}$, $\hat{b}$ and $\hat{c}$ are three unit vectors mutually inclined to each other at an angle θ. $\hat{d}$ is also a unit vector such that $\hat{d}\cdot\hat{a} = \hat{d}\cdot\hat{b} = \hat{d}\cdot\hat{c} = \cos\alpha$. Then match the entries for θ given in column-I with their corresponding values for α given in column-II.

यदि $\hat{a}$, $\hat{b}$ तथा $\hat{c}$ तीन इकाई सदिश इस प्रकार है कि वे परस्पर एक दूसरे से θ कोण बनाते है यदि $\hat{d}$ की एक इकाई सदिश इस प्रकार से है कि $\hat{d}\cdot\hat{a} = \hat{d}\cdot\hat{b} = \hat{d}\cdot\hat{c} = \cos\alpha$,तो स्तम्भ–I में θ को स्तम्भ–II में α से मिलान कीजिए।

Column-I (θ)	**Column-II** (α)
A. $\frac{\pi}{2}$	p. $\frac{\pi}{4}$
B. $\frac{\pi}{3}$	q. $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$
C. $\cos^{-1}\left(\frac{1}{4}\right)$	r. $\cos^{-1}\sqrt{\frac{2}{3}}$
D. $\cos^{-1}\left(\frac{5}{8}\right)$	s. $\frac{\pi}{6}$

Ans. (A) Q (B) R (C) P (D) R

Sol. $\hat{d} = k\left(\hat{a}+\hat{b}+\hat{c}\right)$ _____(i)

$\hat{d}\cdot\hat{a} = k\left(\hat{a}\cdot\hat{a}+\hat{a}\cdot\hat{b}+\hat{a}\cdot\hat{c}\right)$

$\Rightarrow \cos\alpha = k(1 + 2\cos\theta)$ _____(ii)

dot with $\hat{d}$

$\hat{d}\cdot\hat{d} = k\left(\hat{d}\cdot\hat{a}+\hat{d}\cdot\hat{b}+\hat{d}\cdot\hat{c}\right)$

$1 = k(3\cos\alpha)$ _____(iii)

(ii) + (iii)

$|3\cos^2\alpha = 1 + 2\cos\theta|$

45. **Match the statements in Column-I** with those in **Column-II.** **[IIT-JEE-2010,]**

Column-I		**Column-II**	
(A)	A line from the origin meets the lines	(p)	– 4

$\frac{x-2}{1}=\frac{y-1}{-2}=\frac{z+1}{1}$ and $\frac{x-\frac{8}{3}}{2}=\frac{y+3}{-1}=\frac{z-1}{1}$

at P and Q respectively. If length PQ = d, then d^2 is

(B) The values of x satisfying (q) 0

$\tan^{-1}(x+3)-\tan^{-1}(x-3)=\sin^{-1}\left(\frac{3}{5}\right)$ are

(C) Non-zero vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ satisfy $\vec{a}\cdot\vec{b}=0$, (r) 4

$(\vec{b}-\vec{a}).(\vec{b}+\vec{c})=0$ and $2|\vec{b}+\vec{c}|=|\vec{b}-\vec{a}|$. If $\vec{a}=\mu\vec{b}+4\vec{c}$ then possible value of μ are

(D) Let f be the function on $[-\pi, \pi]$ given by (s) 5

$f(0)=9$ and $f(x)=\frac{\sin\left(\frac{9x}{2}\right)}{\sin\left(\frac{x}{2}\right)}$ for $x\neq 0$. The value

of $\frac{2}{\pi}\int_{-\pi}^{\pi} f(x)dx$ is

(t) 6

कॉलम **-I** में दिए गए वक्तव्यों का कॉलम **-II** में दिए वक्तव्यों से सुमेल करें।

कॉलम -I		**कॉलम -II**
(A)	मूल बिन्दु से खींची गयी एक सरल रेखा, सरल रेखाओं $\frac{x-2}{1}=\frac{y-1}{-2}=\frac{z+1}{1}$ तथा $\frac{x-\frac{8}{3}}{2}=\frac{y+3}{-1}=\frac{z-1}{1}$ को क्रमशः P एवं Q पर काटती है। यदि लम्बाई PQ = d हो तो d^2 है–	(p) – 4
(B)	जंद $\tan^{-1}(x+3)-\tan^{-1}(x-3)=\sin^{-1}\left(\frac{3}{5}\right)$ को संतुष्ट करने वाले ग के मान हैं–	(q) 0
(C)	अशुन्य सदिश $\vec{a}$, $\vec{b}$ तथा $\vec{c}$, $\vec{a}\cdot\vec{b}=0$, $(\vec{b}-\vec{a}).(\vec{b}+\vec{c})=0$ तथा $2\|\vec{b}+\vec{c}\|=\|\vec{b}-\vec{a}\|$ को संतुष्ट करते हैं। यदि $\vec{a}=\mu\vec{b}+4\vec{c}$ तो μ का/के के संभावित मान है/हैं–	(r) 4
(D)	मानाकि $[-\pi, \pi]$ पर फलन $f; f(0)=9$ तथा $f(x)=\frac{\sin\left(\frac{9x}{2}\right)}{\sin\left(\frac{x}{2}\right)}$ यग $\neq 0$ द्वारा परिभाषित है। तो $\frac{2}{\pi}\int_{-\pi}^{\pi} f(x)dx$ का मान है–	(s) 5
		(t) 6

Ans. **(A) → (t), (B) → (p, r), (C) → (q,s) (D) → (r)**

Sol. **(A)** Let the line through origin is $\frac{x}{\lambda}=\frac{y}{\mu}=\frac{z}{1}$

$\Rightarrow x = \lambda z, y = \mu z$ .....(1)

To find point of intersection of line (1) and line $\frac{x-2}{1} = \frac{y-1}{-2} = \frac{z+1}{1}$ .....(2)

we have $\frac{\lambda z - 2}{1} = \frac{\mu z - 1}{-2} = z + 1$

$\Rightarrow z = \frac{3}{\lambda - 1} = \frac{-1}{\mu + 2}$

$\Rightarrow \lambda + 3\mu + 5 = 0$ .....(3)

To find point of intersection of line (1) and line $\frac{x - \frac{8}{3}}{2} = \frac{y+3}{-1} = \frac{z-1}{1}$ .....(4)

we have $\frac{\lambda z - \frac{8}{3}}{2} = \frac{\mu z + 3}{-1} = \frac{z-1}{1}$

$\Rightarrow z = \frac{2}{3(\lambda - 2)} = \frac{-2}{\mu + 1}$

$\Rightarrow 3\lambda + \mu = 5$ .....(5)

Solving (3) and (5), $\lambda = \frac{5}{2}$ and $\mu = -\frac{5}{2}$

$\therefore$ $z = 2, x = 5, y = -5$ for point P

and $z = \frac{4}{3}, x = \frac{10}{3}, y = -\frac{10}{3}$ for point Q

$\therefore PQ^2 = \frac{4}{9} + \frac{25}{9} + \frac{25}{9} = 6$

(B) $\tan^{-1}(x+3) - \tan^{-1}(x-3) = \sin^{-1}(3/5)$

$\Rightarrow \tan^{-1}\left(\frac{x+3-x+3}{1+x^2-9}\right) = \tan^{-1}(3/4)$

$\Rightarrow \frac{6}{x^2-8} = \frac{3}{4} \Rightarrow x^2 = 16$

$\therefore x = \pm 4$

(C) Since $\vec{a} \cdot \vec{b} = 0$

$\therefore$ Let $\vec{b} = \lambda_1 \hat{i}, \vec{a} = \lambda_2 \hat{j}$

Now $2|\vec{b} + \vec{c}| = |\vec{b} - \vec{a}|$ & $\vec{a} = \mu\vec{b} + 4\vec{c}$ $\Rightarrow$ $2\left|\lambda_1\hat{i} + \frac{\lambda_2\hat{j} - \lambda_1\mu\vec{b}}{4}\right| = |\lambda_1\hat{i} - \lambda_2\hat{j}|$

$|\lambda_1(4-\mu)\hat{i} + \lambda_2\hat{j}| = 2|\lambda_1\hat{i} + \lambda_2\hat{j}|$

squaring

$\lambda_1^2(4-\mu)^2 + \lambda_2^2 = 4\lambda_1^2 + 4\lambda_2^2$ $\Rightarrow$ $3\lambda_2^2 = (12 + \mu^2 - 8\mu)\lambda_1^2$ .....(1)

Also $(\vec{b} - \vec{a}).(\vec{b} + \vec{c}) = 0$

$\Rightarrow \quad (\lambda_1\hat{i}-\lambda_2\hat{j})\cdot\left(\lambda_1\hat{i}+\frac{\lambda_2\hat{j}-\lambda_1\mu\hat{i}}{4}\right)=0 \quad \Rightarrow \quad \frac{\lambda_1^2(4-\mu)-\lambda_2^2}{4}=0$

$\Rightarrow \quad \lambda_2^2=\lambda_1^2(4-\mu)$ .....(2)

from (1) & (2)

$12+\mu^2-8\mu=12-3\mu$

$\Rightarrow \quad \mu^2-5\mu=0 \quad \Rightarrow \quad \mu=0,5\ (\mu=5 \text{ reject})$

(D) $I=\frac{2}{\pi}\int_{-\pi}^{\pi}\frac{\sin\frac{9x}{2}}{\sin\frac{x}{2}}dx=\frac{4}{\pi}\int_0^{\pi}\frac{\sin\frac{9x}{2}\cos\frac{x}{2}}{\sin\frac{x}{2}\cos\frac{x}{2}}dx=\frac{8}{\pi}\int_0^{\pi}\frac{\sin\frac{9x}{2}\cos\frac{x}{2}}{\sin x}dx$

$I=\frac{4}{\pi}\int_0^{\pi}\frac{\sin 5x+\sin 4x}{\sin x}dx$ .....(i)

$(\text{using}\int_0^b f(x)dx=\int_0^b f(a+b-x)dx)$

$=\frac{4}{\pi}\int_0^{\pi}\frac{\sin 5x-\sin 4x}{\sin x}dx$ .....(ii)

Add (i) and (ii)

$I=\frac{4}{\pi}\int_0^{\pi}\frac{\sin 5x}{\sin x}dx$

Consider

$I_k-I_{k-2}=\frac{4}{\pi}\int_0^{\pi}\frac{\sin kx-\sin(k-2)x}{\sin x}dx=\frac{8}{\pi}\int_0^{\pi}\frac{\cos(k-1)x\sin x}{\sin x}dx$

$I_k=I_{k-2}$

so $I_5=I_3 \quad \Rightarrow \quad I_5=I_1=\frac{4}{\pi}\int_0^{\pi}dx=4$

Aliter

Let $I=\frac{2}{\pi}\int_{-\pi}^{\pi}\frac{\sin(9x/2)}{\sin(x/2)}dx$

$I=\frac{4}{\pi}\int_0^{\pi}\frac{\sin(9x/2)}{\sin(x/2)}dx$ .....(1) ($\because$ f(x) is even function)

$I=\frac{4}{\pi}\int_0^{\pi}\frac{\cos(9x/2)}{\cos(x/2)}dx$ .....(2)

$(\text{using}\int_0^b f(x)dx=\int_0^b f(a+b-x)dx)$

Add (1) & (2)

$$I = \frac{4}{\pi}\int_0^{\pi} \frac{\sin 5x}{2\sin(x/2)\cos(x/2)}\,dx = \frac{4}{\pi}\int_0^{\pi} \frac{\sin 5x}{\sin x}\,dx$$

$$\Rightarrow I = \frac{8}{\pi}\int_0^{\pi/2} \frac{\sin 5x}{\sin x}\,dx$$

$$\Rightarrow I = \frac{8}{\pi}\int_0^{\pi/2} \left(\frac{16\sin^5 x - 20\sin^3 x + 5\sin x}{\sin x}\right)dx$$

$$\Rightarrow I = \frac{8}{\pi}\int_0^{\pi/2} (16\sin^4 x - 20\sin^2 x + 5)\,dx$$

$$\Rightarrow I = \frac{8}{\pi}\left[16x\frac{3\times1\times\pi}{4\times2\times2} - 20\times\frac{1\times\pi}{2\times2} + \frac{5\pi}{2}\right]$$

$$\Rightarrow I = \frac{8}{\pi}\left[3\pi - 5\pi + \frac{5\pi}{2}\right]$$

$$\Rightarrow I = 4$$

Hindi (A) माना मूल बिन्दु से जाने वाली रेखा $\frac{x}{\lambda} = \frac{y}{\mu} = \frac{z}{1}$

$\Rightarrow x = \lambda z, y = \mu z$ .....(1)

रेखा (1) और रेखा $\frac{x-2}{1} = \frac{y-1}{-2} = \frac{z+1}{1}$ के प्रतिच्छेदन बिन्दु के लिए .....(2)

$$\frac{\lambda z - 2}{1} = \frac{\mu z - 1}{-2} = z + 1$$

$$\Rightarrow z = \frac{3}{\lambda - 1} = \frac{-1}{\mu + 2}$$

$\Rightarrow \lambda + 3\mu + 5 = 0$ .....(3)

रेखा (1) और रेखा $\frac{x - \frac{8}{3}}{2} = \frac{y+3}{-1} = \frac{z-1}{1}$ के प्रतिच्छेदन के लिए .....(4)

$$\frac{\lambda z - \frac{8}{3}}{2} = \frac{\mu z + 3}{-1} = \frac{z-1}{1}$$

$$\Rightarrow z = \frac{2}{3(\lambda - 2)} = \frac{-2}{\mu + 1}$$

$\Rightarrow 3\lambda + \mu = 5$ .....(5)

(3) और (5) से, $\lambda = \frac{5}{2}$ और $\mu = -\frac{5}{2}$

$\therefore$ बिन्दु P के लिए $z = 2, x = 5, y = -5$

और बिन्दु Q के लिए $z = \frac{4}{3}, x = \frac{10}{3}, y = -\frac{10}{3}$

$\therefore \quad PQ^2 = \frac{4}{9} + \frac{25}{9} + \frac{25}{9} = 6$

(B) $\tan^{-1}(x+3) - \tan^{-1}(x-3) = \sin^{-1}(3/5)$

$\Rightarrow \tan^{-1}\left(\frac{x+3-x+3}{1+x^2-9}\right) = \tan^{-1}(3/4)$

$\Rightarrow \frac{6}{x^2-8} = \frac{3}{4} \Rightarrow x^2 = 16$

$\therefore x = \pm 4$

(C) चूंकि $\vec{a}\cdot\vec{b} = 0$

$\therefore$ माना $\vec{b} = \lambda_1\hat{i}$, $\vec{a} = \lambda_2\hat{j}$

अब $2|\vec{b} + \vec{c}| = |\vec{b} - \vec{a}|$ और $\vec{a} = \mu\vec{b} + 4\vec{c}$

$\Rightarrow \quad 2\left|\lambda_1\hat{i} + \frac{\lambda_2\hat{j} - \lambda_1\mu\vec{b}}{4}\right| = |\lambda_1\hat{i} - \lambda_2\hat{j}|$

$|\lambda_1(4-\mu)\hat{i} + \lambda_2\hat{j}| = 2|\lambda_1\hat{i} + \lambda_2\hat{j}|$

वर्ग करने पर

$\lambda_1^2(4-\mu)^2 + \lambda_2^2 = 4\lambda_1^2 + 4\lambda_2^2$

$\Rightarrow \quad 3\lambda_2^2 = (12 + \mu^2 - 8\mu)\lambda_1^2$ .....(1)

तथा $(\vec{b} - \vec{a})\cdot(\vec{b} + \vec{c}) = 0$

$\Rightarrow \quad (\lambda_1\hat{i} - \lambda_2\hat{j})\cdot\left(\lambda_1\hat{i} + \frac{\lambda_2\hat{j} - \lambda_1\mu\hat{i}}{4}\right) = 0$

$\Rightarrow \quad \frac{\lambda_1^2(4-\mu) - \lambda_2^2}{4} = 0$

$\Rightarrow \quad \lambda_2^2 = \lambda_1^2(4-\mu)$ .....(2)

; 1 द्वय (2) से

$12 + \mu^2 - 8\mu = 12 - 3\mu$

$\Rightarrow \quad \mu^2 - 5\mu = 0 \quad \Rightarrow \quad \mu = 0, 5$ ($\mu = 5$ reject)

(D) $I = \frac{2}{\pi}\int_{-\pi}^{\pi}\frac{\sin\frac{9x}{2}}{\sin\frac{x}{2}}dx = \frac{4}{\pi}\int_0^{\pi}\frac{\sin\frac{9x}{2}\cos\frac{x}{2}}{\sin\frac{x}{2}\cos\frac{x}{2}}dx = \frac{8}{\pi}\int_0^{\pi}\frac{\sin\frac{9x}{2}\cos\frac{x}{2}}{\sin x}dx$

$$I = \frac{4}{\pi}\int_0^{\pi} \frac{\sin 5x + \sin 4x}{\sin x}\,dx \ldots\ldots(i)$$

$$\left(\int_0^b f(x)dx = \int_0^b f(a+b-x)dx \text{ के उपयोग से}\right)$$

$$= \frac{4}{\pi}\int_0^{\pi} \frac{\sin 5x - \sin 4x}{\sin x}\,dx \ldots\ldots(ii)$$

; पद्धऔर (ii) को जोड़ने पर

$$I = \frac{4}{\pi}\int_0^{\pi} \frac{\sin 5x}{\sin x}\,dx$$

मानाकि

$$I_k - I_{k-2} = \frac{4}{\pi}\int_0^{\pi} \frac{\sin kx - \sin(k-2)x}{\sin x}dx = \frac{8}{\pi}\int_0^{\pi} \frac{\cos(k-1)x \sin x}{\sin x}dx$$

$$I_k = I_{k-2}$$

अतः $\quad I_5 = I_3 \Rightarrow \quad I_5 = I_1 = \frac{4}{\pi}\int_0^{\pi} dx = 4$

वैकल्पिक

माना $I = \frac{2}{\pi}\int_{-\pi}^{\pi} \frac{\sin(9x/2)}{\sin(x/2)}\,dx$

$$I = \frac{4}{\pi}\int_0^{\pi} \frac{\sin(9x/2)}{\sin(x/2)}dx \quad \ldots\ldots(1) \qquad (\because\ f(x) \text{ सम फलन है})$$

$$I = \frac{4}{\pi}\int_0^{\pi} \frac{\cos(9x/2)}{\cos(x/2)}dx \quad \ldots\ldots(2)$$

$$\left(\int_0^b f(x)dx = \int_0^b f(a+b-x)dx \text{ के उपयोग से}\right)$$

; 1द्धव (2) को जोड़ने पर

$$I = \frac{4}{\pi}\int_0^{\pi} \frac{\sin 5x}{2\sin(x/2)\cos(x/2)}dx = \frac{4}{\pi}\int_0^{\pi} \frac{\sin 5x}{\sin x}dx$$

$$\Rightarrow I = \frac{8}{\pi}\int_0^{\pi/2} \frac{\sin 5x}{\sin x}dx$$

$$\Rightarrow\ I = \frac{8}{\pi}\int_0^{\pi/2} \left(\frac{16\sin^5 x - 20\sin^3 x + 5\sin x}{\sin x}\right)dx$$

$$\Rightarrow I = \frac{8}{\pi}\int_0^{\pi/2}(16\sin^4 x - 20\sin^2 x + 5)\,dx$$

$$\Rightarrow I = \frac{8}{\pi}\left[16x\frac{3\times1\times\pi}{4\times2\times2} - 20\times\frac{1\times\pi}{2\times2} + \frac{5\pi}{2}\right]$$

$$\Rightarrow I = \frac{8}{\pi}\left[3\pi - 5\pi + \frac{5\pi}{2}\right]$$

$$\Rightarrow I = 4$$

46. Match the following column :

Column-I	**Column-II**
A. $\left\|\int_0^1 (y^2-1)\,dy\right\| + \left\|\int_0^1 (1-y^2)\,dy\right\|$	p. 2
B. A point (α, β, γ) lies on the plane $x + y + z = 2$. The value of γ such that the vector $\vec{a} = \alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$ satisfies $\hat{k}\times(\hat{k}\times\vec{a}) = \vec{0}$	q. $\left\|\int_{-1}^0 \sqrt{1+x}\,dx\right\| + \left\|\int_0^1 \sqrt{1-x}\,dx\right\|$
C. In a triangle ABC, if $\cos A\cos B + \sin A\sin B\sin C = 1$, then the value of $\sin C$ is	r. $\frac{4}{3}$
D. The set of values of a for which the lines $x + y = \|a\|$, $ax - y = 4$ intersect in the region $x > 0, y > 0$, is the interval (a_0, ∞). Then the value of a_0 is	s. 1

निम्न column को सुमेलित कीजिए :

Column-I	**Column-II**
A. $\left\|\int_0^1 (y^2-1)\,dy\right\| + \left\|\int_0^1 (1-y^2)\,dy\right\|$	p. 2
B. एक बिन्दु (α, β, γ) तल $x + y + z = 2$ पर स्थित है γ का मान होगा इस प्रकार कि $\vec{a} = \alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$, $\hat{k}\times(\hat{k}\times\vec{a}) = \vec{0}$ को सतुष्ट करता है	q. $\left\|\int_{-1}^0 \sqrt{1+x}\,dx\right\| + \left\|\int_0^1 \sqrt{1-x}\,dx\right\|$
C. ΔABC में यदि $\cos A\cos B + \sin A\sin B\sin C = 1$, तो $\sin C$ का मान होगा	r. $\frac{4}{3}$
D. ` के मानों का समुच्चय जिसके लिये रेखाऐं $x + y = \|a\|$, $ax - y = 4$	s. 1

क्षेत्र $x > 0, y > 0$ में प्रतिछेद करती हैं, अन्तराल (a_0, ∞)

है तो a_0 का मान होगा

Ans. (A) Q, R (B) P (C) S (D) P

Sol. (A) Solving the two equations, say i.e.,

$x + y = |a|$ and $ax - y = 1$, we get

$x = \frac{|a| + 1}{a + 1} > 0$ and $y = \frac{|a| - 1}{a + 1} > 0$

when $a + 1 > 0$; we get $a > 1$

$\therefore \quad a_0 = 1$

(B) We have, $\vec{a} = \alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$

$\Rightarrow \vec{a}.\hat{k} = \gamma$

Now, $\hat{k} \times (\hat{k} \times \vec{a}) = (\hat{k}.\vec{a})\hat{k} - (\hat{k}.\hat{k})\vec{a}$

$= \gamma\hat{k} - (\alpha\hat{i} + \beta\hat{j} + \gamma\hat{k})$

$\Rightarrow \alpha\hat{i} + \beta\hat{j} = 0$

$\Rightarrow \alpha = \beta = 0$

Also $\alpha + \beta + \gamma = 2$

$\Rightarrow \quad g = 2.$

(C) $\left|\int_0^1 \left(1 - y^2\right) dy\right| + \left|\int_0^1 \left(y^2 - 1\right) dy\right|$

$= 2\int_0^1 (1 - y^2) dy = \frac{4}{3}$

Also $\left|\int_0^1 \sqrt{1 - x}\, dx\right| + \left|\int_{-1}^0 \sqrt{1 + x}\, dx\right|$

$= 2\int_0^1 \sqrt{1 - x}\, dx = \frac{4}{3}$

(D) $\sin A \sin B \sin C + \cos A \cos B$

$\leq \sin A \sin B + \cos A \cos B = \cos(A - B)$

$\Rightarrow \quad \cos(A - B) \geq 1$

$\Rightarrow \quad \cos(A - B) = 1$

$\Rightarrow \quad \sin C = 1.$

47. Match the following column :

Column-I	**Column-II**
A. Let P be the plane passing through the point (2, 1, – 1) and	p. $\frac{\sqrt{5}}{3}$

perpendicular to the line of intersection of the planes $2x + y - z = 3$ and $x + 2y + z = 2$. Then the distance from the point $\left(\sqrt{3}, 2, 2\right)$ to the plane P is

B. If $\lim\limits_{n\to\infty} \sum\limits_{i=1}^{n} \tan^{-1}\left(\frac{1}{2i^2}\right) = t$, then tan t is — q. 1

C. The sides a, b, c of a triangle ABC are in A.P. If $\cos\theta_1 = \frac{a}{b+c}$, $\cos\theta_2 = \frac{b}{a+c}$, $\cos\theta_3 = \frac{c}{a+b}$, then $\tan^2\frac{\theta_1}{2} + \tan^2\frac{\theta_3}{2} =$ — r. $\frac{2}{3}$

D. Let L be the line passing through the point (0, 1, 0) and perpendicular to the plane $x + 2y + 2z = 0$. Then the distance from the point (0, 0, 0) to the line L is — s. 0

निम्न column को सुमेलित कीजिए :

Column-I	**Column-II**
A. माना P एक तल है जो बिन्दु (2, 1, – 1) से गुजरता है एवं तल $2x + y - z = 3$ तथा $x + 2y + z = 2$ की प्रतिच्छेदी रेखा से गुजरता है तो बिन्दु $\left(\sqrt{3}, 2, 2\right)$ की तल P से दूरी बराबर है	p. $\frac{\sqrt{5}}{3}$
B. यदि $\lim\limits_{n\to\infty} \sum\limits_{i=1}^{n} \tan^{-1}\left(\frac{1}{2i^2}\right) = t$, तो tan t का मान होगा	q. 1
C. यदि त्रिभुज ABC की भुजाऐं a, b, c सं. श्रे. में हो तथा $\cos\theta_1 = \frac{a}{b+c}$, $\cos\theta_2 = \frac{b}{a+c}$, $\cos\theta_3 = \frac{c}{a+b}$, तो $\tan^2\frac{\theta_1}{2} + \tan^2\frac{\theta_3}{2} =$	r. $\frac{2}{3}$
D. माना L एक रेखा है जो बिन्दु (0, 1, 0) से गुजरती है तथा तल ग 2ल 2 ञ 0 के लम्बवत् है तो बिन्दु (0, 0, 0) की रेखा L से दूरी होगी	s. 0

Ans. (A) Q (B) Q (C) R (D) P

Sol. (B) $\sum\limits_{i=1}^{\infty} \tan^{-1}\left(\frac{1}{2i^2}\right) = t$.

$$\Rightarrow \sum_{i=1}^{\infty} \tan^{-1}\left(\frac{2}{4i^2 - 1 + 1}\right)$$

$$= \sum_{i=1}^{\infty} \tan^{-1}\left\{\frac{(2i+1)-(2i-1)}{1+(2i-1)(2i+1)}\right\}$$

$= (\tan^{-1}3 - \tan^{-1}1)+(\tan^{-1}5 - \tan^{-1}3) +....$

$+ \{(\tan^{-1}(2n+1) - \tan^{-1}(2n-1))$

$\therefore \quad t = \lim_{n\to\infty} (\tan^{-1}(2n+1) - \tan^{-1}1)$

$$= \lim_{n\to\infty} \tan^{-1}\left(\frac{2n}{1+2n+1}\right) = \frac{\pi}{4}$$

$\therefore \quad \tan t = 1.$

(C) We have, $\cos\theta_1 = \dfrac{1-\tan^2\frac{\theta_1}{2}}{1+\tan^2\frac{\theta_1}{2}} = \dfrac{a}{b+c}$

$$\Rightarrow \quad \tan^2\left(\frac{\theta_1}{2}\right) = \frac{b+c-a}{b+c+a}$$

Also, $\cos\theta_3 = \dfrac{1-\tan^2\frac{\theta_3}{2}}{1+\tan^2\frac{\theta_3}{2}} = \dfrac{c}{a+b}$

$$\Rightarrow \quad \tan^2\frac{\theta_3}{2} = \frac{a+b-c}{a+b+c}$$

$$\Rightarrow \quad \tan^2\frac{\theta_1}{2} + \tan^2\frac{\theta_3}{2}$$

$$= \frac{2b}{a+b+c} = \frac{2b}{3b} = \frac{2}{3}$$

{as, a, b, c are in AP $\Rightarrow$ 2b = a + c}

(D) Line through (0, 1, 0) and perpendicular to plane x + 2y + 2z = 0 is given by

$$\frac{x-0}{1} = \frac{y-1}{2} = \frac{z-0}{2} = r$$

$\therefore$ P(r, 2r + 1, 2r) be the foot of perpendicular on the straight line then

r . 1 + (2r + 1) . 2 + (2r) . 2 = 0

$$\Rightarrow \quad r = -\frac{2}{9}$$

$$\therefore \quad P\left(-\frac{2}{9}, \frac{5}{9}, -\frac{4}{9}\right)$$

$\therefore$ Required perpendicular distance

$$= \sqrt{\frac{4+25+16}{81}} = \frac{\sqrt{5}}{3} \text{ unit.}$$